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• Second Edition •

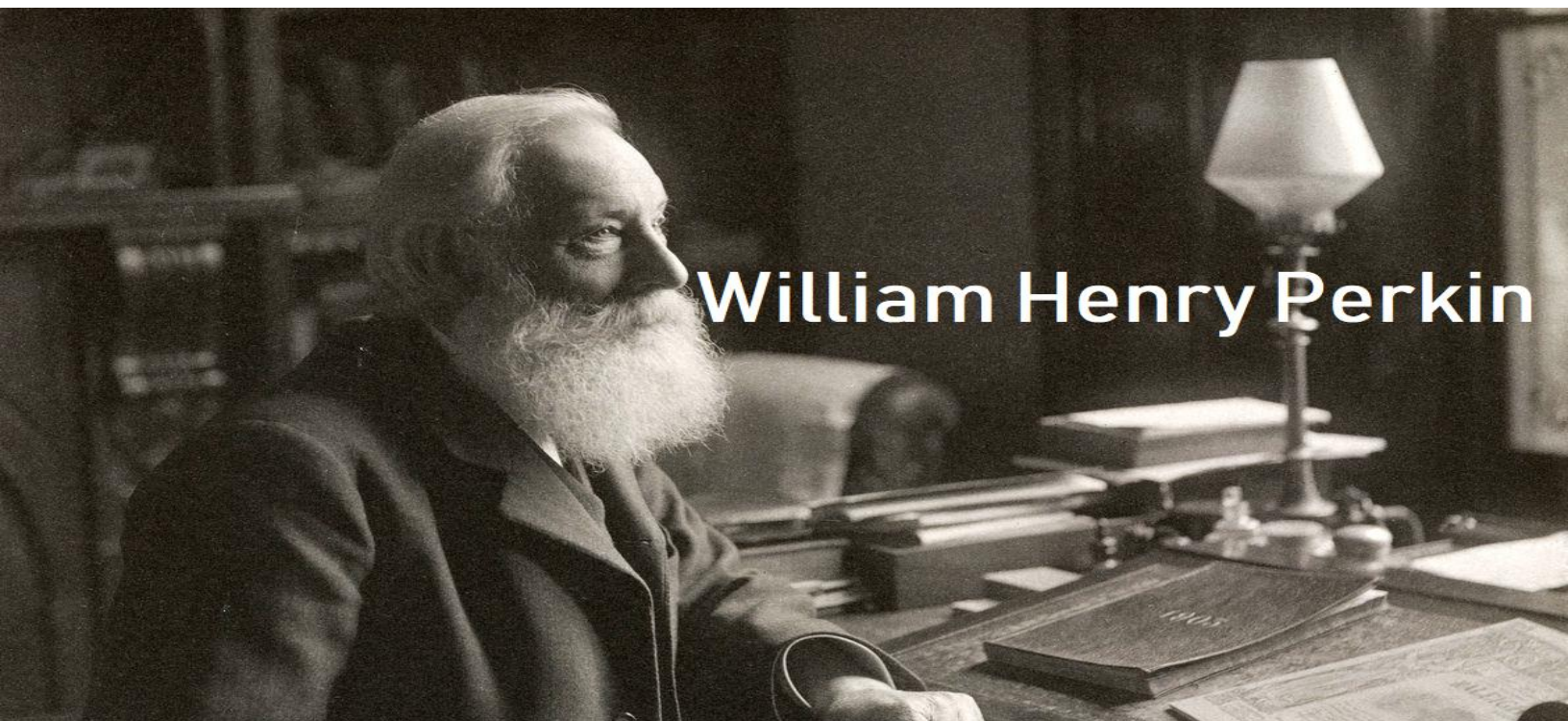
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TEST 1

READING PASSAGE 1



William Henry Perkin

The man who invented **synthetic dyes** Henry Perkin was born on March 12, 1838, in London, England.

As a boy, Perkin's **curiosity** prompted early interests in the arts, sciences, photography, and engineering. But it was a chance **stumbling upon a run-down**, yet **functional**, laboratory in his late grandfather's home that **solidified** the young man's **enthusiasm** for chemistry.

synthetic = artificial, man-made. *sin'θetk*
dye = color, tint, pigment. *daɪ*

curiosity = the desire to know about something. # *apathy* *kjʊəri'psəti*

prompt = to make someone decide to do something. (= encourage, stimulate, motivate) *prɒmpt*

stumble on/across/upon something = come across = to find or discover something by chance and unexpectedly. *'stʌmbəl ɒn ə'kros ə'pɒn 'sʌmθɪŋ*

run-down = in bad condition, badly maintained. *rʌn - daʊn*

functional = useful, handy, practical *'fʌŋkʃənəl*
solidify = strengthen, to make an agreement, plan, attitude etc more definite and less likely to change. *sə'lɪdɪfaɪ*

enthusiasm = interest, passion # *apathy* *ɪn'θju:ziæzəm*

As a student at the City of London School, Perkin became **immersed in** the study of chemistry. His talent and **devotion** to the subject were **perceived** by his teacher, Thomas Hall, who encouraged him to **attend** a series of lectures given by the **eminent** scientist Michael Faraday at the Royal Institution. Those speeches **fired** the young chemist's **enthusiasm** further, and he later went on to **attend** the Royal College of Chemistry, which he succeeded in entering in 1853, at the age of 15.

At the time of Perkin's **enrolment**, the Royal College of Chemistry was headed by the **noted** German chemist August Wilhelm Hofmann. Perkin's scientific **gifts** soon **caught** Hofmann's **attention** and, within two years, he became Hofmann's youngest assistant. Not long after that, Perkin made the scientific **breakthrough** that would bring him both **fame** and **fortune**.

At the time, **quinine** was the only **viable** medical **treatment** for **malaria**. The drug is **derived from** the **bark** of the cinchona tree, **native** to South America, and by 1856 demand for the drug was **surpassing** the available supply. Thus, when Hofmann made some passing comments about the **desirability** of a **synthetic substitute** for quinine, it was unsurprising that his star pupil **was moved** to **take up** the challenge.

During his vacation in 1856, Perkin spent his time in the laboratory on the top floor of his family's house. He was

immersed in = become completely involved in an activity *ɪ'mɜːst ɪn*

devotion= commitment = dedication = the strong love that you show when you pay a lot of attention to someone or something.

dɪ'vəʊʃən

perceive= recognize, see, identify, notice.

pə'siːv

attend= join, be present, go to *ə'tend*

eminent= famous, prominent *'emɪnənt*

fired= inspire, excite, arouse *faɪəd*

enrolment= the process of arranging to join a school, university, course etc. *ɪn'reʊlmənt*

head= control, rule, lead, supervise. *hed*

noted= well-known, famous, eminent *nəʊtɪd*

gift= talent, genius, ability. *ɡɪft*

catch sb attention= to make someone notice you, especially because you want to speak to them or you need their help *kætʃ 'sʌmbədɪ ə'tenʃən*

breakthrough= advance, innovation, revolution, new idea, invention. *'breɪkθruː*

fame= reputation, recognition, celebrity

obscurity *feɪm*

fortune= wealth, riches, opulence, prosperity

#poverty *'fɔːtʃən*

quinine= a drug used for treating fevers, especially malaria *'kwɪniːn*

viable= feasible, possible, successful *'vaɪəbəl*

treatment= cure, medicine, therapy *'triːtmənt*

malaria= a disease that is common in hot countries and that you get when a type of mosquito bites you *mə'leəriə*

derive from= originate, develop, come from *dɪ'raɪv frəm*

bark= the outer covering of a tree *bɑːk*

native to= existing naturally in a place *'neɪtɪv tə*

surpass= transcend, exceed, go beyond *sə'pɑːs*

desirability= something that is desirable is worth having or doing *dɪ'zaɪərəbəl*

substitute= alternate, replacement *'sʌbstɪtjuːt*

be moved= to make someone feel strong emotions *bi muːvd*

take up= start, accept, engage in *teɪk ʌp*

attempting to **manufacture** quinine from aniline, an inexpensive and readily available **coal tar** waste product. Despite his best efforts, however, he did not **end up with** **quinine**. Instead, he produced a **mysterious** dark **sludge**. Luckily, Perkin's scientific training and **nature** **prompted** him to investigate the **substance** further. **Incorporating** potassium dichromate and alcohol into the aniline at various stages of the experimental process, he finally produced a **deep** purple **solution**. And, proving the truth of the famous scientist Louis Pasteur's words 'chance **favours** only the prepared mind', Perkin saw the potential of his unexpected find.

Historically, **textile** dyes were made from such natural sources as plants and animal **excretions**. Some of these, such as the glandular **mucus** of snails, were difficult to obtain and **outrageously** expensive. Indeed, the purple colour **extracted from** a snail was once so costly that in society at the time only the rich could **afford** it. Further, natural dyes tended to be **muddy** in **hue** and **fade** quickly. It was against this **backdrop** that Perkin's discovery was made.

Perkin quickly **grasped** that his purple **solution** could be used to colour **fabric**, thus making it the world's first **synthetic dye**. Realising the importance of this **breakthrough**, he **lost no time in patenting** it. But perhaps the most fascinating of all Perkin's reactions to his find was his nearly **instant recognition** that the new **dye** had **commercial** possibilities.

attempt= try, endeavor, make an effort ə'tempt
manufacture= produce, make, create. mænʃə'fæktʃə
coal tar= a thick black sticky liquid made by heating coal without air kəʊl tɑː
end up with= to be in a particular situation, state, or place after a series of events, especially when you did not plan it end ʌp wɪð
mysterious= strange, unexplained, enigmatic mɪ'stəriəs
sludge= soft thick mud, especially at the bottom of a liquid slʌdʒ
nature= character, personality 'neɪtʃ
incorporate= merge, combine, include #divide ɪn'kɔːpəreɪt
deep= a deep colour is dark and strong #light diːp
solution= mixture, liquid, blend sə'ljuːʃən
favour= help, support, back, approve 'feɪvə

textile= fabric, cloth, material 'tekstaɪl
excretion= the waste material that people or animals get rid of from their bodies. ɪk'skriːʃən
mucus= a thick liquid produced in parts of your body such as your nose 'mjuːkəs
outrageously= very shocking and extremely unfair or offensive aʊt'reɪdʒəs
extract something from something= to carefully remove a substance from something which contains it, using a machine, chemical process etc. ɪk'strækt 'sʌmθɪŋ frəm 'sʌmθɪŋ
afford= pay for, manage to pay for, have enough money. ə'fɔːd
muddy= unclear, murky #clear 'mʌdi
hue= tone, tint, color, shade hjuː
fade= lighten, lose color #darken feɪd
backdrop= the conditions or situation in which something happens. 'bækdɒp

grasp= understand, comprehend. grɑːsp
fabric= material, clothe, textile. 'fæbrɪk
lost no time in doing smt= to do something immediately lɒst nəʊ taɪm ɪn 'duːɪŋ 'sʌmθɪŋ
patent= to obtain a special document giving you the right to make or sell a new invention or product. 'pætnɪt
instant= immediate, instantaneous, prompt 'ɪnstənt
recognition= detection, perception, identification rekəɡ' nɪʃən
commercial= money-making, profitable, saleable kə'mɜːʃəl

Perkin originally named his **dye** Tyrian Purple, but it later became commonly known as mauve (from the French for the plant used to make the colour violet). He asked advice of Scottish **dye** works owner Robert Pullar, who **assured** him that **manufacturing** the **dye** would be well worth it if the colour remained fast (i.e. would not **fade**) and the cost was **relatively** low. So, over the **fierce objections** of his mentor Hofmann, he left college to **give birth to** the modern chemical industry.

With the help of his father and brother, Perkin **set up** a factory not far from London. **Utilising** the cheap and **plentiful coal tar** that was an almost unlimited by product of London's gas street lighting, the **dye** works began producing the world's first synthetically dyed material in 1857. The company received a **commercial boost** from the **Empress** Eugenie of France, when she decided the new colour **flattered** her. Very soon, mauve was the necessary **shade** for all the fashionable ladies in that country.

Not to be outdone, England's Queen Victoria also appeared in public wearing a mauve **gown**, thus making it **all the rage** in England as well. The **dye** was **bold** and fast, and the public **clamoured for** more. Perkin **went back to the drawing board**.

Although Perkin's **fame** was **achieved** and **fortune assured** by his first discovery, the chemist continued his research. Among other dyes he developed and introduced were aniline red (1859) and aniline black (1863) and, in the late 1860s, Perkin's green. It is important to note that Perkin's **synthetic dye** discoveries had outcomes far beyond the **merely** decorative. The dyes also became **vital** to medical research in many ways. For instance, they were used to **stain** previously **invisible** microbes and bacteria, allowing researchers to identify such bacilli as tuberculosis, cholera, and anthrax. Artificial dyes continue to play a **crucial** role today. And, in what would have been particularly **pleasing to** Perkin, their current use is in the search for a vaccine against malaria.

assure= ensure, promise, confirm, guarantee. ə'ʃʊə
relatively= something that is relatively small, easy etc is fairly small, easy etc compared to other things 'relətɪvli
fierce= strong, intense, severe #gentle frɛs
objection= opposition, doubt, concern əb'dʒekʃən
mentor= adviser, tutor, guide 'mento:
give birth to= the time when something new starts to exist grɪv bɜ:θ tə

set up= establish, start, organize set ʌp
utilise= use, apply, exploit, make use of. 'ju:telaɪz
plentiful=more than enough in quantity 'plentɪfəl
boost= increase, improvement, enhancement bu:st
empress= queen, ruler, monarch, king 'emprɪs
flatter= to make someone look as attractive as they can. 'flætə
shade= hue, tint, color ʃeɪd

not to be outdone = in order not to let someone else do better or seem better than you nɒt tə bi aʊt'dʌn
gown= a long dress that a woman wears on formal occasions gaʊn
be all the rage= to be very popular or fashionable. bi ɔ:l ðə reɪdʒ
bold=very strong or bright so that you notice them bəʊld
clamour for= to demand something loudly. 'klæmə fe
go back to the drawing board= if you go back to the drawing board, you start again with a completely new plan or idea, after the one you tried before has failed. ɡəʊ bæk tə ðə 'drɔ:ɪŋ bɔ:d
achieve= attain, get, reach ə'tʃi:v
merely= only, just, simply. 'miəli
vital= very important, crucial, necessary, central. 'vaɪl
stain= mark, spot steɪn
crucial= vital, very important, central. 'kru:ʃəl
pleasing to= giving pleasure, enjoyment, or satisfaction 'pli:zɪŋ tə

READING PASSAGE 2

Is anybody out there?

The Search for **Extra-terrestrial** Intelligence

The question of whether we are alone in the Universe has **haunted** humanity for centuries, but we may now stand **poised on the brink** of the answer to that question, as we search for radio signals from other intelligent **civilisations**. This search, often known by the acronym SETI (search for extra-terrestrial intelligence), is a difficult one. Although groups around the world have been searching **intermittently** for three decades, it is only now that we have reached the level of technology where we can make a **determined** attempt to search all nearby stars for any sign of life.

A

extra-terrestrial= relating to things that exist outside the earth. 'ekstrə tə'restriəl

haunt= to cause problems for someone over a long period of time haʊnt

poised on the brink/edge of something = completely ready to do something or for something to happen, when it is likely to happen soon. pɔɪzd 'ɒn ðə brɪŋk/ edʒ əv 'sʌmθɪŋ

civilisation= a society that is well organized and developed, used especially about a particular society in a particular place or at a particular time sɪvəl-aɪ'zeɪʃən

intermittently= sporadically= stopping and starting often and for short periods. ɪntə'mɪntli

determined= having a strong desire to do something, so that you will not let anyone stop you. dɪ'tɜ:mɪnd

The **primary** reason for the search is basic **curiosity** - the same **curiosity** about the natural world that drives all **pure science**. We want to know whether we are alone in the Universe. We want to know whether life **evolves** naturally if given the right conditions, or whether there is something very special about the Earth to have **fostered** the variety of **life forms** that we see around us on the planet. The simple **detection** of a radio signal will be **sufficient** to answer this most basic of all questions. In this sense, SETI is another **cog in the machinery** of **pure science** which is continually **pushing out** the horizon of our knowledge. However, there are other reasons for being interested in whether life exists elsewhere. For example, we have had **civilisation** on Earth for perhaps only a few thousand years, and the **threats** of nuclear war and pollution over the last few decades have told us that our **survival** may be **tenuous**. Will we **last** another two thousand years or will we **wipe ourselves out**? Since the **lifetime** of a planet like ours is several billion years, we can **expect** that, if other civilisations do survive in our galaxy, their ages will range from zero to several billion years. Thus any other **civilisation** that we hear from is likely to be far older, on average, than ourselves. The **mere** existence of such a **civilisation** will tell us that long-term survival is possible, and gives us some cause for **optimism**. It is even possible that the older **civilisation** may pass on the benefits of their experience in **dealing with threats** to **survival** such as nuclear war and global pollution, and other threats that we haven't yet discovered.

B

In discussing whether we are alone, most SETI scientists **adopt** two **ground rules**. First, UFOs (Unidentified Flying Objects) are generally ignored since most scientists don't **consider** the **evidence** for them to be strong enough to **bear** serious **consideration** (although it is also important to keep an

primary= main, chief, key, principal, most important. 'praɪməri
curiosity= the desire to know about something. kjuəri'ɒsəti
drive= push, impel, urge draɪv
pure science= a science depending on deductions from demonstrated truths, such as mathematics or logic, or studied without regard to practical applications. pjʊə'saɪəns
evolve= develop, change, grow, progress ɪ'vɒlv
foster= promote, encourage. 'fɒstə
life form= any living thing laɪf fɔ:m
detection= discovery, finding. dɪ'tekʃən
sufficient= enough, plenty, adequate sə'fɪʃənt
a cog in the machine/wheel= someone who only has a small unimportant job in a large organization. ə'kɒg ɪn ðə mə'ʃi:n/ 'wi:l
push smt out= to produce large quantities of something pʊʃ 'sʌmθɪŋ 'aʊt
threat= danger, risk, menace θret
survival= existence, being #death sə'vaɪvəl
tenuous= uncertain, weak, vague 'tenjuəs
last= survive, live, endure lɑ:st
wipe something out= to destroy, remove, or get rid of something completely. waɪp 'sʌmθɪŋ aʊt
lifetime= lifespan, duration, lifecycle 'laɪftaɪm
expect= anticipate, await, wait for ɪk'spekt
mere= used to emphasize that something which is small or not extreme has a big effect or is important mɪə
optimism= a tendency to believe that good things will always happen
 #pessimism 'ɒptəmɪzəm
pass smt on (to somebody)= to give something to somebody else, especially after receiving it or using it yourself pɑ:s 'sʌmθɪŋ 'ɒn (tu 'sʌmbədi)
deal with= to take the necessary action, especially in order to solve a problem di:l wɪð
adopt= accept, assume, approve ə'dɒpt
ground rules= the basic rules or principles on which future actions or behaviour should be based. graʊnd ru:lz
consider= think, believe, contemplate kən'sɪdə
evidence= proof, testimony, fact 'evɪdəns
bear= to be responsible for or accept something beə
consideration= thought, contemplation, deliberation kən,sɪdə'reɪʃən

open mind in case any really **convincing evidence** **emerges** in the future). Second, we make a very **conservative assumption** that we are looking for a **life form** that is pretty well like us, since if it differs **radically** from us we may well not recognise it as a **life form**, quite apart from whether we are able to communicate with it. In other words, the **life form** we are looking for may well have two green heads and seven fingers, but it will nevertheless **resemble** us in that it should communicate with **its fellows**, be interested in the Universe, live on a planet **orbiting** a star like our Sun, and perhaps most restrictively, have a chemistry, like us, based on carbon and water.

C

Even when we make these **assumptions**, our understanding of other **life forms** is still **severely** limited. We do not even know, for example, how many stars have planets, and we certainly do not know how likely it is that life will **arise** naturally, given the right conditions. However, when we look at the 100 billion stars in our galaxy (the Milky Way), and 100 billion galaxies in the **observable** Universe, it seems **inconceivable** that at least one of these planets does not have a **life form** on it; in fact, the best **educated guess** we can make, using the little that we do know about the conditions for carbon-based life, leads us to estimate that perhaps one in 100,000 stars might have a life-bearing planet **orbiting** it. That means that our nearest neighbours are perhaps 100 light years away, which is almost next door in **astronomical** terms.

D

An **alien civilisation** could choose many different ways of sending information across the galaxy, but many of these either **require** too much energy, or else

an open mind= if you have an open mind, you deliberately do not make a decision or form a definite opinion about something.

ən 'əʊpən maɪnd

convincing= persuasive, believable, credible. kən 'vɪnsɪŋ

emerge= arise, appear, occur, develop. ɪ'mɜːdʒ

conservative = traditional, old-fashioned, conventional. kən 'sɜːvətɪv

assumption= supposition, guess, hypothesis, theory, belief. ə'sʌmpʃən

radically= very, completely, totally, drastically. 'rædɪkli

resemble= look like, be similar to #differ somebody's fellows= people that you work with, study with, or who are in the same situation as you. ɪ'zembəl

sb's fellows=people that you work with, study with, or who are in the same situation as you 'səm,bə:di 'feləʊz

orbit= to travel in a curved path around a much larger object such as the earth, the sun etc. 'ɔːbrɪt

severely= strictly, cruelly, harshly #gently sə'veɪəli

arise= happen, occur, start, appear. ə'raɪz

observable= noticeable, visible, apparent, obvious əb'zɜːvəbəl

inconceivable= unthinkable, unimaginable, unbelievable. ɪnkən'si:vəbəl

educated guess= a guess that is likely to be correct because it is based on some knowledge 'edʒuketɪd ges

estimate= calculate approximately. 'estəmət

astronomical= relating to the scientific study of the stars. æstrə'nomɪkəl

alien= in stories, a creature from another world 'eɪliən

require= need, involve, entail. ɪ'kwaɪə

are **severely attenuated** while **traversing** the **vast** distances across the galaxy. It turns out that, for a given amount of **transmitted** power, radio waves in the **frequency** range 1000 to 3000 MHz travel the greatest distance, and so all searches **to date** have **concentrated on** looking for radio waves in this **frequency** range. So far there have been a number of searches by **various** groups around the world, including Australian searches using the **radio telescope** at Parkes, New South Wales. Until now there have not been any **detections** from the few hundred stars which have been searched. The **scale** of the searches has been increased **dramatically** since 1992, when the US **Congress** voted NASA \$10 million per year for ten years to **conduct** a **thorough** search for **extra-terrestrial** life. Much of the money in this project is being spent on developing the special **hardware** needed to search many **frequencies** at once. The project has two parts. One part is a **targeted** search using the world's largest **radio telescopes**, the American-operated telescope in Arecibo, Puerto Rico and the French telescope in Nancy in France. This part of the project is searching the nearest 1000 likely stars with high **sensitivity** for signals in the **frequency** range 1000 to 3000 MHz. The other part of the project is an undirected search which is **monitoring** all of space with a lower **sensitivity**, using the smaller **antennas** of NASA's Deep Space Network.

E

There is **considerable debate** over how we should react if we detect a signal from an **alien civilisation**. Everybody agrees that we should not reply immediately. Quite apart from the **impracticality** of sending a reply over such large distances **at short notice**, it raises a **host of ethical** questions that would

severely= harshly, strictly. sə'viəli
attenuate= weaken, reduce, decrease ə'tenjuet
traverse= pass through= to move across, over, or through something, especially an area of land or water. 'trævɜ:s
vast= huge, massive, immense vɑ:st
transmit= transfer, communicate, spread trænz'mit
frequency= the number of radio waves, sound waves etc that pass any point per second. 'fri:kwensi
to date= up to now. tu deɪt
concentrate on= focus on, think, deliberate, contemplate. 'kɒnsəntreɪt ɒn
various= numerous, many, countless. 'veəriəs
radio telescope= a piece of equipment that collects radio waves from space and is used to find stars and other objects in space. 'reɪdiəʊ 'teləskəʊp
detection= discovery, finding, recognition dɪ'tekʃən
scale= size, level, range, degree skeɪl
dramatically= radically, noticeably, considerably, significantly. drə'mætɪkli
Congress= the group of people elected to make laws in the US, consisting of the senate and the house of representatives 'kɒŋɡres
conduct= do, perform, carry out kən'dʌkt
thorough= detailed, full, comprehensive 'θʌrə
hardware= the machinery and equipment that is needed to do something. 'hɑ:dweə
target= aim, point, direct 'tɑ:ɡɪt
sensitivity= the ability to react to very small changes in light, heat, movement etc sense'trɪvəti
monitor= check, observe, watch, supervise 'mɒnɪtə
antenna= a wire rod etc used for receiving radio and television signals æn'tenə

considerable= substantial, large, significant, huge. kən'sɪdərəbəl
debate= argument, discussion. dɪ'beɪt
impracticality= not sensible or possible for practical reasons ɪm'præktɪkəl
at short notice= if you do something at short notice, you do not have very much time to prepare for it ət ʃɔ:t 'nəʊtɪs
a host of= a large number of people or things ə'hoʊst ɒv
ethical= moral, principled #unethical 'eθɪkəl

have to be **addressed** by the global community before any reply could be sent. Would the human race **face** the **culture shock** if faced with a **superior** and much older **civilisation**? Luckily, there is no **urgency** about this. The stars being searched are hundreds of light years away, so it takes hundreds of years for their signal to reach us, and a further few hundred years for our reply to reach them. It's not important, then, if there's a delay of a few years, or decades, while the human race debates the question of whether to reply, and perhaps carefully **drafts** a reply.

address a problem/question/issue etc= if you address a problem, you start trying to solve it əˈdres ə ˈprɒbləm ˈkwɛstʃən ˈɪʃuː etˈsetrə
face= tackle, confront, handle feɪs
culture shock= the feeling of being confused or anxious that you get when you visit a foreign country or a place that is very different from the one you are used to ˈkʌltʃə ʃɒk
superior= someone who has a higher rank or position than you, especially in a job suːˈpiəriə
urgency= very important and needing to be dealt with immediately ˈɜːdʒənsi
draft= to write a plan, letter, report etc that will need to be changed before it is in its finished form draːft

READING PASSAGE 3

The history of the tortoise

If you **go back** far enough, everything lived in the

sea. At various points in **evolutionary** history, **enterprising** individuals within many different animal groups **moved out** onto the land, sometimes even to the most **parched** deserts, taking their own private seawater with them in blood and **cellular** fluids. In addition to the **reptiles**, birds, **mammals** and insects which we see all around us, other groups that have succeeded out of water include scorpions, snails, crustaceans such as woodlice and land crabs, millipedes and centipedes, spiders and various worms. And we mustn't forget the plants, without whose **prior**

go back= to have existed since a time in the past gəʊ bæk

evolutionary= relating to the way in which plants and animals develop and change gradually over a long period of time. iːvəˈluːʃənəri

enterprising= having the ability to think of new activities or ideas and make them work ˈentəpraɪzɪŋ

move out= leave, depart, set off #move in muːv aʊt

parched= dry, arid, waterless. pɑːtʃt

cellular= consisting of or relating to the cells of plants or animals ˈseljələ

reptile= a type of animal, such as a snake or lizard, whose body temperature changes according to the temperature around it, and that usually lays eggs to have babies. ˈreptail

mammal= a type of animal that drinks milk from its mother's body when it is young. humans, dogs, and whales are mammals. ˈmæməl

prior = previous, past, earlier, preceding. ˈpraɪə

invasion of the land none of the other **migrations** could have happened.

Moving from water to land involved a major redesign of every aspect of life, including breathing and

reproduction. Nevertheless, a good number of

thoroughgoing land animals later turned around,

abandoned their **hard-earned** **terrestrial** **re-tooling**,

and returned to the water again. Seals have only gone

part way back. They show us what the **intermediates**

might have been like, **on the way to extreme cases**

such as whales and dugongs. Whales (including the

small whales we call dolphins) and dugongs, with their

close cousins the manatees, ceased to be land

creatures altogether and **reverted to** the full marine

habits of their **remote** **ancestors**. They don't even

come **ashore** to **breed**. They do, however, still breathe

air, having never developed anything **equivalent to**

the **gills** of their earlier marine **incarnation**. Turtles

went back to the sea a very long time ago and, like all

vertebrate **returnees** to the water, they breathe air.

However, they are, **in one respect**, less fully given

invasion= the arrival in a place of a lot of people or things, often where they are not wanted in'veɪʒən

migration= when birds or animals travel regularly from one part of the world to another maɪ'greɪʃən

reproduction= breeding, procreation. ri:prə'dʌkʃən

thoroughgoing= complete, thorough, absolute. θə'reʊ'gəʊɪŋ

turn around= if a business, department etc that is not successful turns around, or if someone turns it around, it starts to be successful tɜ:n ə'raʊnd

abandon= discard, dump, throw away. ə'bændən

hard-earned= earned or achieved after a lot of effort 'hɑ:d'ɜ:n

terrestrial= living on or relating to land rather than water. tə'restriəl

re-tool= to organize something in a new way ri:'tu:l

intermediate= in-between, halfway, middle intə'mi:diət

be on the way to sth= to be close to doing something bi 'ɒn ðə 'wei tu 'sʌmθɪŋ

extreme case= very unusual and severe or serious ɪk'stri:m keɪs

cease to do smt= to stop doing something or stop happening 'si:s tu du: 'sʌmθɪŋ

creature= animal, living thing, organism 'kri:tʃə

revert to somebody/something= go back to= to change back to a situation that existed in the past. ri'vɜ:t tə 'sʌmbədi/ 'sʌmθɪŋ

remote= distant, isolated, faraway. ri'məʊt

ancestor= an animal that lived in the past, that modern animals have developed from 'ænsəstə

ashore= on or towards the shore of a lake, river, sea etc. ə'ʃɔ:

breed= reproduce, procreate, have a baby brɪd

equivalent to= having the same value, purpose, job etc as a person or thing of a different kind. ɪ'kwɪvələnt tə

gill= one of the organs on the sides of a fish through which it breathes ɡɪl

incarnation= the state of living in the form of a particular person or animal. according to some religions, people have several different incarnations. ɪnka:'neɪʃən

vertebrate= a living creature that has a backbone 'vɜ:təbreɪt

returnee= a person who returns to their own country after living in another country ri,tʃ:'ni:

in one respect/in some respects etc= used to say that something is true in one way, in some ways etc ɪn wʌn rɪ'spekt ɪn səm rɪ'spekts et'setə

back to the water than whales or dugongs, for turtles still **lay** their eggs on beaches.

There is evidence that all modern turtles are **descended from a terrestrial ancestor** which lived before most of the dinosaurs. There are two key **fossils** called Proganochelys quenstedti and Palaeochersis talampayensis dating from early dinosaur times, which appear to be close to the **ancestry** of all modern turtles and tortoises. You might wonder how we can tell whether **fossil** animals lived on land or in water, especially if only **fragments** are found. Sometimes it's obvious. Ichthyosaurs were reptilian **contemporaries** of the dinosaurs, with **fins** and **streamlined** bodies. The **fossils** look like dolphins and they surely lived like dolphins, in the water. With turtles it is a little less obvious. One way to tell is by measuring the bones of their **forelimbs**. Walter Joyce and Jacques Gauthier, at Yale University, **obtained** three measurements in these particular bones of 71 **species** of living turtles and tortoises. They used a kind of triangular **graph** paper to **plot** the three measurements against one another. All the land tortoise **species** formed a tight **cluster of** points in the upper part of the triangle; all the water turtles **cluster** in the lower part of the triangular **graph**. There was no **overlap**, except when they added some **species** that spend time both in water and on land. Sure enough, these **amphibious species** show up on the triangular graph **approximately** half way between the 'wet **cluster**' of sea turtles and the 'dry **cluster**' of land tortoises. The next step was to **determine** where the **fossils** fell. The bones of P quenstedti and JR talampayensis **leave us in no doubt**. Their points on

lay= if a bird, insect etc lays eggs, it produces them from its body *leɪ*

descend from= to have developed from something that existed in the past *dɪ'send frəm*

fossil= an animal or plant that lived many thousands of years ago and that has been preserved, or the shape of one of these animals or plants that has been preserved in rock *'fɒsəl*

ancestry= the members of your family who lived a long time ago *'ænsəstri*

fragment= piece, part, portion *#whole 'frægmənt*

contemporary= someone who lived or was in a particular place at the same time as someone else. *kən'tempərəri*

fin= one of the thin body parts that a fish uses to swim *fɪn*

streamlined= designed or arranged in a way that makes movement easier through air or water *'stri:mlaɪnd*

forelimb = one of the two front legs of an animal with four legs. *'fɔ:leg*

obtain= get, gain, attain, acquire. *əb'teɪn*

species= group, type, class, kind, sort *'spi:ʃi:z*

graph= chart, diagram, table. *grɑ:f*

plot= to draw marks or a line to represent facts, numbers etc. *plɒt*

cluster of something= bunch, group, collection *'klʌstə əv 'sʌmθɪŋ*

cluster= gather, group, assemble, collect *'klʌstə*

overlap= the amount by which two things or activities cover the same area *əʊvə'leɪp*

amphibious= able to live both on land and in water. *æm'fɪbiəs*

approximately= about, around, roughly, almost, nearly. *ə'prɒksəmətli*

determine= decide, find out, verify. *dɪ'tɜ:mɪn*

leave no/little doubt (that)=make people sure or almost sure about something *li:v nəʊ/ 'lɪtl daʊt (ðæt)*

the **graph** are right **in the thick of** the dry cluster. Both these **fossils** were dry-land tortoises. They come from the **era** before our turtles returned to the water. You might think, therefore, that modern land tortoises have probably stayed on land ever since those early **terrestrial** times, as most **mammals** did after a few of them went back to the sea. But apparently not. If you draw out the **family tree** of all modern turtles and tortoises, nearly all the branches are **aquatic**. Today's land tortoises **constitute** a single branch, deeply nested among branches consisting of **aquatic** turtles. This suggests that modern land tortoises have not stayed on land continuously since the time of P. quenstedti and P. talampayensis. Rather, their **ancestors** were among those who **went back** to the water, and they then re-**emerged** back onto the land in (**relatively**) more recent times.

Tortoises therefore **represent** a **remarkable** double return. **In common with** all mammals, reptiles and birds, their **remote ancestors** were marine fish and before that various more or less worm-like creatures stretching back, still in the sea, to the **primeval** bacteria. Later **ancestors** lived on land and stayed there for a very large number of **generations**. Later **ancestors** still **evolved** back into the water and became sea turtles. And finally they returned yet again to the land as tortoises, some of which now live in the driest of deserts.

in the thick of= involved in the busiest, most active, most dangerous etc part of a situation *ɪn ðə θɪk əv*

era= period, age, time. *ˈɪərə*

family tree= a drawing that gives the names of all the members of a family over a long period of time, and shows how they are related to each other *ˈfæməli triː*

aquatic= living or growing in water. *əˈkwætɪk*

constitute = form, establish, create, set up *ˈkɒnstɪtjuːt*

emerge= arise, appear, occur, develop *ɪˈmɜːdʒ*

relatively= something that is relatively small, easy etc is fairly small, easy etc compared to other things *ˈrelatɪvli*

represent= signify, denote, stand for *reprɪˈzent*

remarkable= noteworthy, notable, significant. *ɪˈmɑːkəbəl*

in common with= in the same way as someone or something else *ɪn ˈkɒmən wɪð*

primeval= ancient, prehistoric #modern *praɪˈmiːvəl*

generation= all the members of a family of about the same age *dʒenəˈreɪʃən*

evolve= change, grow, develop, progress *ɪˈvɒlv*

Nếu học được một lượng từ vựng lớn thì các bạn sẽ không phải quan tâm đến tip này hay trick kia khi làm bài thi IELTS Reading. Mình tin là có những bạn 1 tuần đọc liên tục được 2 cuốn Boost your vocabulary, thậm chí là hơn. Truyện dài mấy trăm trang mà nhiều bạn có thể đọc xong trong 1 đêm, còn 1 cuốn Boost your vocabulary là khá mỏng, và lại toàn từ đã được tra sẵn. Vậy nên hãy cố gắng đọc thật nhanh nhé các bạn 😊

Đinh Thắng

TEST 2

READING PASSAGE 1



A

Hearing impairment or other **auditory** function **deficit** in young children can have a major impact on their development of **speech** and communication, resulting in a **detrimental** effect on their ability to learn at school. This is likely to have major **consequences** for the individual and the population as a whole. The New Zealand Ministry of Health has found from research **carried out** over two decades that 6-10% of children in that country are affected by hearing loss.

B

A **preliminary** study in New Zealand has shown that classroom noise presents a **major** concern for teachers and pupils. Modern teaching practices, the organisation of desks in the classroom, poor classroom **acoustics**,

mental/visual/cognitive/hearing etc impairment= a condition in which a part of a person's mind or body is damaged or does not work well. 'mentl/ 'vɪʒuəl/ 'kɒgnətɪv/ 'hiərɪŋ et 'setərə ɪm'peəmənt
auditory = connected with hearing 'ɔ:dətəri
deficit= shortfall, shortage, insufficiency
 #surplus 'defɪsɪt
speech= verbal communication= the ability to speak. spi:tʃ
detrimental= harmful, damaging= causing harm or damage. detre'mentl
consequence= result, outcome, effect. 'kɒnsɛkwəns
carry out= do, perform, conduct. 'kæri 'aʊt

preliminary= initial, opening, primary
 pri'lɪmənəri

major= main, key, chief #minor 'meɪdʒə

acoustics= the shape and size of a room, which affect the way sound is heard in it.
 ə'ku:stɪks

and mechanical means of **ventilation** such as air-conditioning units all **contribute** to the number of children unable to **comprehend** the teacher's voice. Education researchers Nelson and Soli have also suggested that recent trends in learning often involve **collaborative interaction** of **multiple** minds and tools as much as individual **possession** of information. This all **amounts to heightened** activity and noise levels, which have the **potential** to be particularly serious for children experiencing **auditory function deficit**. Noise in classrooms can only **exacerbate** their difficulty in comprehending and processing **verbal** communication with other children and **instructions** from the teacher.

C

Children with **auditory** function **deficit** are potentially **failing to learn** to their maximum **potential** because of noise levels **generated** in classrooms. The effects of noise on the ability of children to learn effectively in **typical** classroom environments are now the subject of increasing concern. The International Institute of Noise Control Engineering (I-INCE), on the advice of the World Health Organization, has established an international **working party**, which includes New Zealand, to **evaluate** noise and **reverberation** control for school rooms.

D

While the **detrimental** effects of noise in classroom situations are not limited to children experiencing **disability**, those with a **disability** that affects their processing of speech and **verbal** communication could

ventilation= the fact of allowing fresh air to enter and move around a room, building, etc.

'ventɪleɪt

contribute to= to help to make something happen kən'trɪbjʊt tə

comprehend= understand, know, figure out.

kəm'prɪ'hend

collaborative= joint, two-away, combined

kə'læbə'reɪtɪv

interaction= communication, contact,

ɪntər'ækʃən

multiple= several, many, various 'mʌltɪpəl

possession= ownership, tenure. pə'zefən

amount to= sum, total, aggregate. ə'maʊnt tə

heighten= increase, intensify, improve,

enhance 'haɪn

potential= possibility, latent, dormant pə'tenʃəl

exacerbate= worsen, aggravate, impair.

ɪg'zæsəbeɪt

verbal= spoken rather than written 'vɜ:bəl

instruction= teaching, training, tuition,

guidance. ɪn'strʌkʃən

fail to do sth= to not succeed in

achieving something 'feɪl tu du: 'sʌmtɪŋ

generate= produce, create, make.

'dʒenəreɪt

typical= usual, normal, standard 'tɪpɪkəl

working party= team, working group,

committee 'wɜ:kɪŋ 'pɑ:ti

evaluate= assess, estimate, value.

ɪ'vælju'eɪt

reverberation= echo= a loud sound that is

heard again and again as it is sent back

from different surfaces ɪ'vɜ:bə'reɪʃən

disability= frailty = debility = a physical or mental condition that makes it difficult for someone to use a part of their body properly, or to learn normally. dɪsə'bɪləti

be extremely **vulnerable**. The **auditory** function **deficits** in question include **hearing impairment**, autistic **spectrum disorders** (ASD) and attention **deficit disorders** (ADD/ADHD).

E

Autism is considered a **neurological** and **genetic** life-long **disorder** that causes **discrepancies** in the way information is processed. This **disorder** is **characterised** by **interlinking** problems with social imagination, social communication and social **interaction**. According to Janzen, this affects the ability to understand and relate in **typical** ways to people, understand events and objects in the environment, and understand or **respond** to **sensory stimuli**. **Autism** does not allow learning or thinking in the same ways as in children who are developing **normally**.

Autistic spectrum disorders often result in **major** difficulties in **comprehending verbal** information and **speech** processing. Those experiencing these **disorders** often find sounds such as crowd noise and the noise **generated** by machinery painful and **distressing**. This is difficult to scientifically **quantify** as such extra-**sensory stimuli** vary greatly from one **autistic** individual to another. But a child who finds any type of noise in their classroom or learning space **intrusive** is likely to be **adversely** affected in their ability to process information.

F

The attention **deficit disorders** are **indicative** of

vulnerable= weak= someone who is vulnerable can be easily harmed or hurt
'vʌlnərəbəl
disorder= a mental or physical illness which prevents part of your body from working properly. dis'ɔ:də
spectrum= range, band, scale 'spektrəm

autism= a mental disorder (=problem) that makes people unable to communicate properly, or to form relationships. 'ɔ:tizəm
neurological= the scientific study of the nervous system and its diseases

njuərə'lɒdʒɪkl
genetic= relating to genes or genetics
dʒə'netɪk

discrepancy in/ between= difference, inconsistency, divergence dɪ'skrepənsɪ ɪn/brɪ'twi:n

characterise= describe, portray, illustrate, depict. 'kærəktəraɪz

interlink= to connect or be connected with something else. ɪntə'ɪŋk

respond to= react = to do something as a reaction to something that has been said or done rɪ'spɒnd tə

extra- (prefix)= outside or beyond 'ekstrə

sensory= relating to or using your senses of sight, hearing, smell, taste, or touch. 'sensəri

stimulus (plural stimuli) = something that makes someone or something move or react. 'stɪmjələs

distressing= stressful = upset, painful. dɪ'stresɪŋ

quantify= measure, calculate, count. 'kwɒntɪfaɪ

vary= differ, diverge, contrast 'veəri

intrusive= disturbing, unpleasant, upset. ɪn'tru:sɪv

adversely= harmfully, badly, negatively. 'ædvɜ:sli

be indicative of something= to be a clear sign that a particular situation exists or that something is likely to be true. bi ɪn'dɪkətɪv əv 'sʌmθɪŋ

neurological and **genetic disorders** and are **characterised** by difficulties with **sustaining** attention, effort and **persistence**, organisation skills and **disinhibition**. Children experiencing these **disorders** find it difficult to **screen out** unimportant information, and **focus** on everything in the environment rather than attending to a **single** activity. **Background** noise in the classroom becomes a major **distraction**, which can affect their ability to **concentrate**.

G

Children experiencing an **auditory** function **deficit** can often find speech and communication very difficult to **isolate** and process when **set against** high levels of **background** noise. These levels come from outside activities that **penetrate** the classroom **structure**, from teaching activities, and other noise **generated** inside, which can be **exacerbated** by room **reverberation**. **Strategies** are needed to **obtain** the **optimum** classroom construction and perhaps a change in classroom culture and methods of teaching. **In particular**, the effects of noisy classrooms and activities on those experiencing **disabilities in the form of auditory** function **deficit** need **thorough** investigation. It is **probable** that many **undiagnosed** children exist in the education system with '**invisible**' **disabilities**. Their needs are less likely to be met than those of children with known **disabilities**.

H

The New Zealand Government has developed a New Zealand Disability **Strategy** and has **embarked on** a **wide-ranging consultation** process. The **strategy**

sustain= maintain, protract, keep up sə'steɪn
persistence= tenacity, diligence, insistence
 pə'sɪstəns
inhibition= a feeling of shyness or embarrassment that stops you doing or saying what you really want #disinhibition ɪnhə'bɪʃən
screen something out/ screen out something= filter out= to remove people or things that are not acceptable or not suitable.
 skri:n 'sʌmθɪŋ aʊt/ skri:n aʊt 'sʌmθɪŋ
background= environment, surroundings, setting. 'bækgraʊnd
distraction= something that stops you paying attention to what you are doing. dɪ'strækʃən
concentrate= focus, think, deliberate, contemplate 'kɒnsəntreɪt

isolate = separate, divorce, insulate. 'aɪsəleɪt
set against= to make someone start to fight or quarrel with another person, especially a person who they had friendly relations with before set ə'geɪnst
penetrate= go through, enter, invade. 'penetreɪt
strategy= plan, scheme, approach. 'strætədʒi
obtain= get, gain, achieve, acquire. əb'teɪn
optimum = best, optimal, ideal, prime.
 'ɒptəməm
in particular= especially, specially, particularly
 ɪn pə'tɪkjələ
in the form of= having the shape, character, style etc of ɪn ðə 'fɔ:m ɒv
thorough= full, detailed, comprehensive 'θʌrə
probable= likely, possible #unlikely 'prɒbəbəl
diagnose= to find out what illness someone has, or what the cause of a fault is, after doing tests, examinations etc. 'daɪəgnəʊz
invisible= hidden, concealed, disguised
 #obvious ɪn'vɪzəbəl

embark on/upon something= to start something, especially something new, difficult, or exciting. ɪm'bɑ:k 'ɒn ə'pɒn 'sʌmθɪŋ
wide-ranging= widespread, comprehensive, thorough. waɪd - 'reɪndʒɪŋ
consultation= discussion, talk, conference, meeting. kɒnsəl'teɪʃən

recognises that people experiencing disability **face** significant **barriers** in achieving a full **quality of life** in areas such as attitude, education, employment and access to services. Objective 3 of the New Zealand Disability **Strategy** is to 'Provide the Best Education for Disabled People' by improving education so that all children, youth learners and adult learners will have equal opportunities to learn and develop within their already existing local school. For a successful education, the learning environment is **vital** significant, so any effort to improve this is likely to be of great benefit to all children, but especially to those with **auditory** function **disabilities**.

I

A number of countries are already in the **process of formulating** their own **standards** for the control and reduction of classroom noise. New Zealand will probably follow their example. The **literature to date** on noise in school rooms appears to focus on the effects on school children **in general**, their teachers and **the hearing impaired**. Only limited attention appears to have been given to those students experiencing the other **disabilities** involving **auditory** function **deficit**. It is **imperative** that the needs of these children are **taken into account** in the **setting** of **appropriate** international **standards** to be **promulgated** in future.

face= confront, meet, encounter. *feɪs*

barrier= difficulty, obstacle, hindrance
'bæriə

quality of life= the level of enjoyment, comfort and health in someone's life:
'kwɒlɪti əv laɪf

vital= crucially, fundamentally, essentially. *'vaɪtli*

process of= a series of things that happen naturally and result in gradual change *'prəʊsɪs*
ɒv

formulate= invent, create, make, plan
'fɔ:mjʊleɪt

standard= criterion, model, pattern, requirement. *'stændəd*

literature on something = all the books, articles etc on a particular subject. *'lɪtrətʃər 'ɒn 'sʌmθɪŋ*

to date= up to now. *tu deɪt*

in general= generally, on the whole, overall, in most cases. *ɪn 'dʒenrəl*

the hearing impaired= people who are not able to hear well *ðə 'hiəriŋ ɪm'peəd*

imperative= urgent, very important, crucial, vital. *ɪm'perətɪv*

take into account = allow for, take into consideration, bear/keep in mind #ignore *'teɪk 'ɪntə ə'kaʊnt*

appropriate= suitable, proper, fitting. *ə'prəʊpri-ət*

promulgate= to spread an idea or belief to as many people as possible. *'prɒmʊlgeɪt*

READING PASSAGE 2

Venus in transit

June 2004 saw the first **passage**, known as a '**transit**', of the planet Venus across the face of the Sun in 122 years. Transits have helped shape our **view** of the whole Universe, as Heather Cooper and Nigel Henbest **explain**

A

On 8 June 2004, more than half the population of the world were **treated** to a rare **astronomical** event. For over six hours, the planet Venus **steadily inched** its way over the surface of the Sun. This '**transit**' of Venus was the first since 6 December 1882. On that **occasion**, the American **astronomer** Professor Simon

passage= movement, motion, travel, crossing 'pæsɪdʒ

transit= the process of moving goods or people from one place to another 'trænsɪt

view= opinion, belief, notion vju:

explain= describe, clarify, give details, make clear. ɪk'spleɪn

treat sb to sth= entertain sb with sth special trɪ:t 'sʌmbədi tu 'sʌmθɪŋ

astronomical= relating to the scientific study of the stars. æstrə'nomɪkəl

steadily= gradually, little by little, inch by inch. 'stedəli

inch= to move very slowly in a particular direction, or to make something do this ɪntʃ

occasion= event, time, circumstance ə'keɪʒən

astronomer= a scientist who studies the stars and planets ə'strɒnəmə

Newcomb led a **party** to South Africa to **observe** the event. They were **based** at a girls' school, where - **it is alleged** - the combined **forces** of three **schoolmistresses** **outperformed** the **professionals** with the **accuracy** of their observations.

B

For centuries, **transits** of Venus have **drawn** explorers and astronomers alike to **the four corners of the globe**. And you can **put it all down to** the **extraordinary polymath** Edmond Halley. In November 1677, Halley observed a **transit** of the **innermost** planet, Mercury, from the **desolate** island of St Helena in the South Pacific. He realised that, from different **latitudes**, the passage of the planet across the Sun's **disc** would appear to differ. By timing the **transit** from two widely-separated locations, teams of **astronomers** could calculate the **parallax** angle - the **apparent** difference in position of an **astronomical** body **due to** a difference in the observer's position. Calculating this angle would allow **astronomers** to measure what was then the **ultimate** goal: the distance of the Earth from the Sun. This distance is known as the **astronomical** unit' or AU.

party= group, team, organization 'pɑ:ti
observe= watch, scrutinize, monitor. əb'zɜ:v
base= to have your main place of work, business etc in a particular place beɪs
it is alleged(that)= to say that something is true or that someone has done something wrong, although it has not been proved. 'ɪt ɪz ə'ledʒd (ðæt)
force= a group of people who have been trained and organized to do a particular job fɔ:s
schoolmistress= a female teacher, especially in a private school (=one that parents pay to send their children to). 'sku:l,mɪstrɪs
outperform= surpass, outdo, do better than. aʊtpə'fɔ:m
professional= showing that someone has been well trained and is good at their work prə'feʃənəl
accuracy=exactness, precision, exactness. 'ækjərəsi

draw somebody to something= attract, appeal, lure, entice. drɔ: 'sʌmbədi tu 'sʌmθɪŋ
the four corners of the earth/world/globe= places or countries that are very far away from each other. ðə fɔ:r 'kɔ:nəz əv ði 'z:θ/'wɜ:ld/ gləʊb
put down to= to think that something is caused by something else 'put daʊn tu:
extraordinary=amazing, special, remarkable. ɪk'strɔ:dənəri
polymath= someone who has a lot of knowledge about many different subjects. 'pɒlɪmæθ
innermost= furthest inside or nearest to the centre. 'ɪnəməʊst
desolate= isolated, uninhabited, wild. 'desələt
latitude= the distance north or south of the equator (=the imaginary line around the middle of the world), measured in degrees. 'lætɪtju:d
disc= a round flat shape or object disk
parallax= the effect by which the position or direction of an object appears to change when the object is seen from different positions 'pærələks
apparent= obvious, clear, noticeable. ə'pærənt
due to= because of, owing to, as a result to dju: tu:
ultimate= eventual, last, final. 'ʌltəmət

C

Halley was aware that the AU was one of the most **fundamental** of all **astronomical** measurements. Johannes Kepler, in the early 17th century, had shown that the distances of the planets from the Sun **governed** their **orbital** speeds, which were easily measurable. But no-one had found a way to calculate **accurate** distances to the planets from the Earth. The **goal** was to measure the AU; then, knowing the **orbital** speeds of all the other planets round the Sun, the **scale** of the Solar System would **fall into place**. However, Halley realised that Mercury was so far away that its **parallax** angle would be very difficult to **determine**. As Venus was closer to the Earth, its **parallax** angle would be larger, and Halley **worked out** that by using Venus it would be possible to measure the Sun's distance to 1 part in 500. But there was a problem: **transits** of Venus, unlike those of Mercury, are rare, **occurring in pairs** **roughly** eight years apart every hundred or so years. **Nevertheless**, he **accurately predicted** that Venus would cross the face of the Sun in both 1761 and 1769 - though he didn't **survive** to see either.

D

Inspired by Halley's suggestion of a way to **pin down** the **scale** of the Solar System, teams of British and French **astronomers** **set out on** expeditions to places as **diverse** as India and Siberia. But things weren't helped by Britain and France being at war. The person

fundamental = basic, major, elementary. fʌndə'mentl
govern= control, rule, regulate 'gʌvən
orbit= the curved path travelled by an object which is moving around another much larger object such as the earth, the sun etc -> orbital (adj) 'ɔ:bɪt
accurate = precise, correct, exact 'ækjərət
goal= objective, aim, purpose, target. geʊl
scale= size, range, extent skeɪl
fall into place= make sense, become clear, take shape. fɔ:l 'ɪntə 'pleɪs
determine= find out, establish, form. dɪ'tɜ:mɪn
work out= to calculate an answer, amount, price etc 'wɜ:k aʊt
occur= happen, take place. ə'kɜ:
in pairs= in groups of two ɪn peəz
roughly= approximately, about, around, nearly. 'rʌfli
nevertheless= but, however, nonetheless, though. nevəðə'les
predict= forecast, foresee, guess. prɪ'dɪkt
survive= live, stay alive, last sə'vaɪv

inspire= motivate, encourage, stimulate. ɪn'spaɪə
pin sb/sth down= to understand something clearly or be able to describe it exactly. pɪn 'sʌmbədi/'sʌmθɪŋ daʊn
set out on a journey/drive/voyage etc = to start a journey, especially a long journey set aʊt 'ɒn ə 'dʒɜ:nɪ/ 'draɪv /'vɔɪdʒ et'set
expedition= journey, trip, tour, voyage. ɪkspə'dɪʃən
diverse= different, varied, assorted. daɪ'vɜ:s

who **deserves** most **sympathy** is the French

astronomer Guillaume Le Gentil.

He was **thwarted** by the fact that the British were

besieging his observation **site** at Pondicherry in India.

Fleeing on a French warship crossing the Indian

Ocean, Le Gentil saw a wonderful **transit** - but the

ship's **pitching and rolling** **ruled out** any **attempt** at

making **accurate** observations. **Undaunted**, he

remained south of **the equator**, keeping himself busy

by studying the islands of Mauritius and Madagascar

before **setting off** to **observe** the next **transit** in the

Philippines. **Ironically** after travelling nearly 50,000

kilometres, his view was **clouded** out at the last

moment, a very **dispiriting** experience.

E

While the early **transit** timings were as **precise** as

instruments would allow, the measurements were

dogged by the 'black drop' effect. When Venus

begins to cross the Sun's **disc**, it looks **smeared** not

circular - which makes it difficult to **establish** timings.

This is **due to diffraction** of light. The second problem

is that Venus **exhibits** a **halo of** light when it is seen

just outside the Sun's **disc**. While this showed

astronomers that Venus was surrounded by a thick

layer of gases **refracting** sunlight around it, both

effects made it impossible to **obtain accurate** timings.

F

But **astronomers laboured** hard to **analyse** the

deserve= be worthy of, ought to have.

di'zɜ:v

sympathy= the feeling of being sorry for someone who is in a bad situation 'sɪmpəθi

thwart= prevent, impede, ruin, frustrate

θwɔ:t

besiege= to surround a city or castle with military force until the people inside let you take control. br'si:dʒ

flee= run away, escape, run off. fli:

pitch and roll= an up and down movement of a ship or an aircraft 'pɪtʃ ənd rɔ:l

rule smt/sb out= preclude, prohibit,

prevent. ru:l 'sʌmθɪŋ/ 'sʌmbədi 'aʊt

attempt= effort, try, endeavor. ə'tempt

undaunted= undeterred= not afraid of continuing to try to do something in spite of difficulties or danger. ʌn'dɔ:ntɪd

the equator= an imaginary line drawn around the middle of the earth that is exactly the same distance from the north pole and the south pole ði i'kwetə

set off= to start to go somewhere set of

ironically= used when talking about a situation in which the opposite of what you expected happens or is true ai'ronɪkli

cloud= to make sth less pleasant or enjoyable klaʊd

dispiriting= disappointing, upsetting, hopeless di'spɪrɪtɪŋ

precise= exact, accurate, specific. pri'saɪs

instrument= tool, gadget, device 'ɪnstɹəmənt

dog= trouble, bother, hassle dɒg

black drop effect= an optical phenomenon visible during a transit of venus and, to a lesser extent, a transit of mercury. blæk drɒp i'fekt

smeared= dirty, muddy, messy smiəd

establish= set up, create, launch. i'stæblɪʃ

diffract= deflection, spreading, diversion -> diffraction (n) di'frækt

exhibit= show, display, present ɪg'zɪbɪt

halo of= a circle of light or something bright. 'heɪləʊ ɒv

refract= if glass or water refracts light, the light changes direction when it passes through the glass or water. ri'frækt

obtain= get, gain, attain, acquire. əb'teɪn

labour= to work hard. 'leɪbə

analyse= investigate, study, examine, scrutinize. 'ænəlaɪz

results of these **expeditions** to **observe** Venus **transits**. Johann Franz Encke, Director of the Berlin Observatory, finally **determined** a value for the AU based on all these **parallax** measurements: 153,340,000 km. Reasonably **accurate** for the time, that is quite close to today's value of 149,597,870 km, **determined** by radar, which has now **superseded** **transits** and all other methods in **accuracy**. The AU is a **cosmic** measuring rod, and the basis of how we **scale** the Universe today. The **parallax** principle can be extended to measure the distances to the stars. If we look at a star in January - when Earth is at one point in its **orbit** - it will seem to be in a different position from where it appears six months later. Knowing the width of Earth's **orbit**, the **parallax shift** lets **astronomers** calculate the distance.

G

June 2004's **transit** of Venus was thus more of an **astronomical spectacle** than a scientifically important event. But such **transits** have **paved the way for** what might prove to be one of the most **vital** **breakthroughs** in the **cosmos** - detecting Earth-sized planets orbiting other stars.

expedition= journey, trip, tour, voyage.
ekspə'diʃən

observe= watch, scrutinize, monitor.
əb'zɜ:v

transit= the process of moving goods or people from one place to another
'trænsɪt

determine= find out, establish, form.
dɪ'tɜ:mɪn

parallax= the effect by which the position or direction of an object appears to change when the object is seen from different positions 'pærələks

accurate = precise, correct, exact
'ækjərət

supersede= replace= if a new idea, product, or method supersedes another one, it becomes used instead because it is more modern or effective. su:pə'si:d

cosmic= vast, huge, immense. #tiny
'kɒzmɪk

scale= to make writing or a picture the right size for a particular purpose skeɪl

extend= make bigger, expand, enlarge.
ɪk'stend

shift = move, change, modification ʃɪft

spectacle= a very impressive show or scene. 'spektəkel

pave the way for= to make a later event or development possible by producing the right conditions. peɪv ðə 'wei fɔ:

vital= very important, crucial, central, necessary. 'vaɪtl

breakthrough= advance, innovation, invention. 'breɪkθru:

cosmos= the whole universe, especially when you think of it as a system 'kɒzmɒs

READING PASSAGE 3

A neuroscientist reveals how to think differently

In the last decade a **revolution** has occurred in the way that scientists think about the brain.

We now know that the decisions humans make can be **traced** to the **firing patterns** of neurons in specific parts of the brain. These discoveries have led to the field known as neuroeconomics, which studies the brain's secrets to success in an economic environment that demands **innovation** and being able to do things differently from **competitors**. A brain that can do this is an **iconoclastic** one. **Briefly**, an **iconoclast** is a person who does something that others say can't be done.

This definition **implies** that **iconoclasts** are different from other people, but more **precisely**, it is their brains that are different in three **distinct** ways: **perception**, fear response, and social intelligence.

revolution= change, development, innovation. revə'lu:ʃən

trace sth (back) to something= to find the origins of when something began or where it came from. treɪs 'sʌmθɪŋ bæk tu 'sʌmθɪŋ

fire= to generate an electrical impulse faɪə
pattern= a regularly repeated arrangement of sounds or words 'pætən

innovation= invention, improvement, advance. ɪnə'veɪʃən

competitor= contestant, participant, rival. kəm'petɪtə

iconoclastic= iconoclastic ideas, opinions, writings etc attack established beliefs and customs. aɪ kənə'klæstɪk

iconoclast= someone who attacks established ideas and customs aɪ kənəklæst
briefly= in a few words, concisely. 'bri:flɪ

imply= suggest, indicate, mean. ɪm'plaɪ

precisely= exactly, accurately, specifically. prɪ'saɪsli

distinct= different, separate, diverse. dɪ'stɪŋkt

perception= view, opinion, assessment. pə'sepʃən

Each of these three functions **utilizes** a different **circuit** in the brain. Naysayers might suggest that the brain is **irrelevant**, that thinking in an original, even **revolutionary**, way is more **a matter of personality** than brain function. But the field of neuroeconomics **was born out of** the **realization** that the physical workings of the brain place limitations on the way we make decisions. By understanding these **constraints**, we begin to understand why some people **march to a different drumbeat**.

The first thing to realize is that the brain **suffers from** limited resources. It has a **fixed** energy **budget**, about the same as a 40 watt light bulb, so it has **evolved** to work as efficiently as possible. This is where most people are **impeded** from being an **iconoclast**. For example, when **confronted with** information streaming from the eyes, the brain will **interpret** this information in the quickest way possible. Thus it will draw on both past experience and any other source of information, such as what other people say, to **make sense of** what it is seeing. This happens all the time. The brain takes **shortcuts** that work so well we are **hardly** ever **aware of** them.

We think our **perceptions** of the world are real, but they are only biological and electrical **rumblings**. **Perception** is not simply a product of what your eyes or ears **transmit** to your brain. More than the

utilize= use, employ, make use of. 'ju:təlaɪz
circuit= a closed system of wires or pipes through which electricity or liquid can flow 'sɜ:kɪt
irrelevant= unrelated, beside the point #relevant. ɪ'reləvənt
revolutionary= new, innovative, groundbreaking revə'lu:ʃənəri
a matter of= only, just. ə 'mætər ɒv
personality= character, nature, trait. pɜ:sə'neɪləti
be born (out) of= existing as a result of a particular situation bi bɔ:n (aʊt) ɒv
realization= understanding, comprehensiion, grasp riələɪ'zeɪʃən
constraint= limitation, restriction, restraint. kən'streɪnt
march to (the beat of) a different drummer= to behave or do things in a manner that does not conform to the standard, prevalent, or popular societal norm. mɑ:tʃ tu (ðə bi:t əv) ə 'dɪfrənt 'drʌmə
suffer from= experience, bear, undergo, tolerate. 'sʌfə frəm
fixed= unchanging, permanent, static. fɪkst
budget= resources, financial plan, funds. 'bʌdʒɪt
evolve= develop, change, grow. ɪ'vɒlv
impede= hinder, prevent, inhibit, block. ɪm'pi:d
be confronted with something= meet, face, encounter, tackle. bi kən'frʌntɪd wɪð 'sʌmθɪŋ
interpret = explain, clarify, translate ɪn'tɜ:pɪt
make (some) sense of sth= understand, comprehend, grasp. 'meɪk səm sens əv 'sʌmθɪŋ
shortcut= a quicker way of doing something. 'ʃɔ:t,kʌt
hardly= barely, only just, almost not 'hɑ:dlɪ
aware of= if you are aware that a situation exists, you realize or know that it exists ə'weər ɒv
rumbling= a series of long low sounds transmit= communicate, transfer, convey. 'rʌmblɪŋ
transmit= communicate, transfer, pass on. trænzmɪt

physical reality of photons or sound waves,

perception is a product of the brain.

Perception is **central** to **iconoclasm**. **Iconoclasts** see things differently to other people. Their brains do not **fall into** efficiency **pitfalls** as much as the average person's brain. **Iconoclasts**, either because they were born that way or through learning, have found ways to **work around** the **perceptual shortcuts** that **plague** most people. **Perception** is not something that is **hardwired** into the brain. It is a learned process, which is both a **curse** and an opportunity for change. The brain **faces** the **fundamental** problem of **interpreting** physical **stimuli** from the senses.

Everything the brain sees, hears, or touches has multiple **interpretations**. The one that is **ultimately** chosen is simply the brain's best **theory**. In technical terms, these **conjectures** have their basis in the **statistical likelihood** of one **interpretation** over another and are heavily influenced by past experience and, importantly for **potential iconoclasts**, what other people say.

The best way to see things differently to other people is to **bombard** the brain with things it has never **encountered** before. **Novelty** releases the **perceptual** process from the **chains** of past experience and forces the brain to make new judgments. Successful **iconoclasts** have an

central= vital, essential, fundamental, crucial, important. 'sentrəl

fall into a trap/pitfall= to make a mistake that many people make fɔ:l 'intə ə 'træp/ 'pɪtfɔ:l

work around sb/sth=to arrange or organize something so that you avoid problems that may stop you from doing something 'wɜ:k ə 'raʊnd 'sʌmbədi/ 'sʌmθɪŋ
perceptual = relating to the ability to become aware of something (to perceive = verb) pə'septʃuəl

plague= trouble, dog, torture. pleɪg
hardwired= if an attitude, way of behaving etc is hard-wired, it is a natural part of a person's character that they are born with and cannot change. hɑ:d 'waɪəd

curse= trouble, plague, burden. kɜ:s

face= meet, encounter, cope with. feɪs

fundamental= basic, primary, elementary. fʌndə'mentl

stimulus (plural stimuli)= something that makes someone or something move or react. 'stɪmjələs

ultimately= finally, eventually, at last. 'ʌltəmətli

theory= general principles and ideas about a subject 'θəri

conjecture= guess, hypothesis, assumption, estimation. kən'dʒektʃə

statistical= numerical, arithmetic. stə'tɪstɪkəl

likelihood= possibility, chance, probability. 'laɪklihʊd

potential= possible, probable, latent, likely. pə'tenʃəl

bombard sb with sth= to do something too often or too much, for example criticizing or questioning someone, or giving too much information. bɒm'bɑ:d 'sʌmbədi wɪð 'sʌmθɪŋ

encounter= meet, face, come across. ɪn'kaʊntə

novelty= innovation, newness, uniqueness. 'nɒvəlti

chain= group, sequence, series, string. tʃeɪn

extraordinary willingness to be **exposed** to what is fresh and different. **Observation** of **iconoclasts** shows that they **embrace novelty** while most people avoid things that are different.

The problem with **novelty**, however, is that it tends to **trigger** the brain's fear system. Fear is a major **impediment** to thinking like an **iconoclast** and **stops** the average person **in his tracks**. There are many types of fear, but the two that **inhibit iconoclastic** thinking and people generally find difficult to **deal with** are fear of uncertainty and fear of public **ridicule**. These may seem like **trivial phobias**. But fear of public speaking, which everyone must do from time to time, **afflicts** one-third of the population. This makes it too common to be considered a **mental disorder**. It is simply a common **variant** of human nature, one which **iconoclasts** do not let **inhibit** their reactions.

Finally, to be successful **iconoclasts**, individuals must **sell** their ideas to other people. This is where social intelligence **comes in**. Social intelligence is the ability to understand and manage people in a business setting. In the last decade there has been an **explosion** of knowledge about the social brain and how the brain works when groups **coordinate** decision making. **Neuroscience** has **revealed** which brain circuits are **responsible** for functions like

extraordinary= uncommon, strange, unusual, odd. ɪk'strɔːdənəri
willingness= enthusiasm, motivation, eagerness 'wɪlɪŋnəs
expose= to make it possible for someone to experience new ideas, ways of life etc ɪk'spəʊz
observation= surveillance, inspection, study, examination. əbzə'veɪʃən
embrace= accept, welcome, adopt. ɪm'breɪs

trigger= activate, generate, prompt. 'trɪɡə
impediment= barrier, block, hindrance, obstacle. ɪm'pedəmənt
stop/halt (dead) in your tracks= to suddenly stop, especially because something has frightened or surprised you stɒp/ hɔːlt (ded) ɪn jə træks
inhibit= hinder, deter, prevent. ɪn'hɪbɪt
deal with= manage, cope with, handle diːl wɪð
ridicule= laughter, mockery, scorn. 'rɪdəkjʊːl
trivial= small, minor, unimportant. 'trɪviəl
phobia= a strong unreasonable fear of something 'feʊbiə
afflict= trouble, bother, upset, distress. ə'flɪkt
mental= relating to the health or state of someone's mind 'mentl
disorder= a mental or physical illness which prevents part of your body from working properly. dɪs'ɔːdə
variant= different, abnormal, irregular. 'veəriənt

sell= to try to make someone accept a new idea or plan, or to become accepted sel
come in= to become fashionable or popular kʌm ɪn
explosion= sudden or quick increase in the number or amount of something ɪk'spləʊʒən
coordinate= collaborate = to organize an activity so that the people involved in it work well together and achieve a good result. kəʊ'ɔːdɪneɪt
neuroscience= the scientific study of the brain 'njʊərəʊs,saɪəns
reveal= tell, disclose, show, bring to light. ri'viːl

understanding what other people think, **empathy**, fairness, and social identity. These brain regions play key roles in whether people **convince** others of their ideas. **Perception** is important in **social cognition** too. The **perception** of someone's enthusiasm, or **reputation**, can make or break a **deal**.

Understanding how **perception** becomes **intertwined with** social decision making shows why successful **iconoclasts** are so rare.

Iconoclasts create new opportunities in every area from **artistic** expression to technology to business. They supply creativity and **innovation** not easily **accomplished** by **committees**. Rules aren't important to them. **Iconoclasts** face **alienation** and failure, but can also be a major **asset** to any organization. It is **crucial** for success in any field to understand how the **iconoclastic** mind works.

empathy= the ability to understand other people's feelings and problems. 'empəθi

convince= persuade, induce, prove..
kən'vɪns

social cognition= concerned with the study of the thought processes, both implicit and explicit, through which humans attain understanding of self, others, and their environment. 'səʊʃəl kɒg'nɪʃən

reputation= name, status, standing.

repjə'teɪʃən

deal= agreement, contract, transaction di:l

intertwine with = connect, link,
interconnect ɪntə'twain wɪð

accomplish= achieve, complete, do

ə'kʌmplɪʃ

committee= group, team, board kə'mɪti

face= cope with, confront, tackle feɪs

alienation= the feeling of not being part of society or a group eɪliə'neɪʃən

major= main, chief, key. 'meɪdʒə

asset= advantage, strength, benefit.

'æset

crucial= vital, central, important, necessary, fundamental. 'kru:ʃəl

Nhiều bạn ngại là đọc cuốn này xong thì làm test không đánh giá đúng nữa? Thật ra câu trả lời là **KHÔNG PHẢI NHƯ VẬY**. Ở đầu sách đã ghi rất rõ là bạn cần phải làm test trước và sau đó thì dùng cuốn này để không phải mất công tra từ điển, cộng thêm với việc học synonym (từ đồng nghĩa) để hiểu đoạn văn nói gì.

Tuy nhiên, với các bạn band Reading đã ở tầm 7.0-8.0 thì cũng không cần phải làm test quá nhiều nữa. Tập trung vào đọc bài cho thật hiểu, đến từ nào không biết thì nhìn sang cột bên phải quyển Boost này để xem nghĩa của từ và lại đọc tiếp. **ĐỌC, ĐỌC, ĐỌC. HIỂU, HIỂU, HIỂU**. Cứ thế đọc mấy cuốn này như đọc báo, **KHÔNG PHẢI LÀM TEST NHIỀU**. Tự khắc điểm sẽ lên 8.0-9.0. Vì nếu học từ vựng mà không hiểu nội dung bài đọc thật sâu thì cũng vô nghĩa.

Hy vọng là sách bộ Boost your vocabulary - Cambridge IELTS này sẽ tiếp tục giúp được nhiều bạn tiết kiệm thời gian & đạt kết quả thật cao trong Reading!

Đình Thắng

TEST 3

READING PASSAGE 1



It is not easy to be **systematic** and **objective**

about language study. Popular **linguistic debate** regularly **deteriorates** into **invective** and **polemic**. Language belongs to everyone, so most people feel they have a right to hold an opinion about it. And when opinions differ, emotions can run high. Arguments can start as easily over **minor** points of **usage** as over major policies of **linguistic** education.

systematic= organized carefully and done thoroughly. sɪstə'mætɪk

objective= based on facts, or making a decision that is based on facts rather than on your feelings or beliefs. əb'dʒektɪv

linguistic= related to language, words, or linguistics lɪŋ'ɡwɪstɪk

debate= argument, discussion, dispute. dɪ'beɪt

deteriorate= worsen, get worse, decline, weaken dɪ'tɪəriəreɪt

invective= rude and insulting words that someone says when they are very angry. ɪn'vektɪv

polemic = a written or spoken statement that strongly criticizes or defends a particular, idea, opinion, or person. pə'lemɪk

minor= small, insignificant, minimal, unimportant #major. 'maɪnə

usage= the way that words are used in a language 'ju:sɪdʒ

Language, moreover, is a very public behaviour, so it is easy for different **usages** to be noted and **criticised**. No part of society or social behaviour is **exempt**: **linguistic** factors influence how we judge **personality**, intelligence, social status, educational standards, job **aptitude**, and many other areas of **identity** and social **survival**. As a result, it is easy to hurt, and to be hurt, when language use is unfeelingly attacked.

In its most general **sense**, prescriptivism is the view that one **variety** of language has an **inherently** higher value than others, and that this ought to be **imposed** on the whole of the speech community. The view is **propounded** especially **in relation to** grammar and vocabulary, and frequently **with reference to** pronunciation. The **variety** which is **favoured**, in this account, is usually a version of the 'standard' written language, especially as **encountered** in literature, or in the formal spoken language which most closely reflects this style. **Adherents** to this **variety** are said to speak or write 'correctly'; **deviations from** it are said to be 'incorrect'!

All the main languages have been studied **prescriptively**, especially in the 18th century approach to the writing of grammars and dictionaries. The aims of these early grammarians were **threefold**: (a) they wanted to **codify** the principles of their languages, to show that there was a system beneath the **apparent chaos** of **usage**, (b) they wanted a means of **settling disputes** over **usage**, and (c) they wanted to **point out** what they felt to be common errors, in order to 'improve' the language. The

criticize= assess, analyse, evaluate.

'krɪtɪsaɪz

exempt= not affected by something =

exclude = except ɪg'zempt

personality= character, nature, trait

pɜːsə'næləti

aptitude= ability, skill, capacity. 'æptɪtjuːd

identity= distinctness, uniqueness,

personality aɪ'dentəti

survival= existence, being, persistence

sə'vaɪvəl

sense= the ability to understand or judge

something sens

a variety of sth= a lot of things of the same type that are different from each other in some way

ə'veəriəti əv 'sʌmθɪŋ

inherently= naturally, essentially, innately #

superficially ɪn'hɪərəntli

impose sth on smb = to force someone to have the same ideas, beliefs etc as you. ɪm'pəʊz

'sʌmθɪŋ 'ɒn 'sʌmbədi

propound= to suggest an idea, explanation etc for other people to consider. prə'paʊnd

in relation to sth= used to talk about something that is connected with or compared with the thing you are talking about ɪn rɪ'leɪʃn tu 'sʌmθɪŋ

in/with reference to sth= used to say what you are writing or talking about ɪn/ wɪθ 'refrəns tu 'sʌmθɪŋ

favour= prefer, choose, support, back. 'feɪvə

encounter= meet, come across, stumble upon.

ɪn'kaʊntə

adherent= supporter, believer, follower əd'hɪərənt

deviation from= a noticeable difference from what is expected or acceptable dɪ'viː'eɪʃn frɒm

prescriptively= describing how the words of a language should be used, rather than describing how it is actually used prɪ'skrɪptɪvli

threefold= three times as much or as many

'θriːfəʊld

codify= to arrange laws, principles, facts etc in a system 'kəʊdɪfaɪ

apparent= evident, clear, obvious, visible.

ə'pærənt

chaos= disorder, mess, anarchy 'keɪ-ɒs

settle a dispute/ lawsuit/ conflict/ argument= to end an argument or solve a disagreement. 'setl ə dɪ'spjuːt/ 'lɔːsuːt/ kən'fɪkt/ 'ɑːgjumənt

point out= show, reveal, indicate pɔɪnt 'aʊt

authoritarian nature of the approach is best **characterised** by its **reliance** on 'rules' of grammar. Some **usages** are '**prescribed**,' to be learnt and followed accurately; others are '**proscribed**,' to be avoided. In this early **period**, there were no half-measures: **usage** was either right or wrong, and it was the task of the grammarian not simply to record alternatives, but to **pronounce** judgement **upon** them.

These **attitudes** are still with us, and they **motivate** a widespread concern that **linguistic** standards should be maintained. Nevertheless, there is an alternative point of view that is concerned less with standards than with the facts of **linguistic usage**. This approach is summarised in the statement that it is the task of the grammarian to describe, not **prescribe** to record the facts of **linguistic diversity**, and not to attempt the impossible tasks of **evaluating** language **variation** or **halting** language change. In the second half of the 18th century, we already find **advocates** of this view, such as Joseph Priestley, whose **Rudiments** of English Grammar (1761) insists that 'the custom of speaking is the original and only just standard of any language! **Linguistic** issues, it is argued, cannot be solved by logic and **legislation**. And this view has become the **tenet** of the modern **linguistic** approach to grammatical analysis. In our own time, the opposition between '**descriptivists**' and '**prescriptivists**' has often become **extreme**, with both sides painting

authoritarian= strictly forcing people to obey a set of rules or laws, especially ones that are wrong or unfair. ɔːθə're'teəriən
characterise= describe, portray, illustrate.

'kærəktəraɪz

reliance= dependence ɪ'laɪəns

prescribe=recommend, propose, advocate.

prɪ'skraɪb

proscribe= prohibit, ban, forbid, disallow.

prəʊ'skraɪb

pronounce on/upon sth= to give a judgment or opinion on something prə'naʊns
'ɒn/ ə'pɒn

attitude= thought, mindset, viewpoint

'ætɪtjuːd

motivate= inspire, encourage, prompt.

'məʊtɪveɪt

diversity= the fact of including many different types of people or things daɪ'vɜːsəti

evaluate= assess, appraise, value. ɪ'væljuːeɪt

variation= difference, distinction, discrepancy. veəri'eɪʃən

halt = prevent, stop, discontinue. hɔːlt

advocate=supporter, backer, promoter, believer 'ædvəkeɪt

rudiments= basics, essential, fundamentals 'ruːdɪmənts

legislation = law, rule, statute. ledʒə'sleɪʃən

tenet = principle, belief, rule, theory. 'tenɪt

descriptivist = someone who believes that books about language should describe how language is really used, rather than giving rules to follow saying what is correct and not correct dɪ'skrip.tɪ.vɪst

prescriptivist= someone who believes that there are correct and wrong ways to use language and that books about language should give rules to follow, rather than describing how language is really used prɪ'skrip.tɪ.vɪst

extreme= severe, intense, acute ɪk'striːm

unreal pictures of the other. Descriptive grammarians have been presented as people who do not care about standards, because of the way they see all forms of **usage** as equally valid. Prescriptive grammarians have been presented as blind **adherents** to a historical tradition. The opposition has even been presented in **quasi**-political terms - of **radical liberalism** vs **elitist conservatism**.

adherent= supporter, believer, follower
 əd'hiərənt
quasi- (*prefix*)= like something else or trying to be something else kweɪzai
radical = new, different (ideas), against what most people think or believe 'rædɪkəl
liberalism= an attitude of respecting and allowing many different types of beliefs or behaviour # conservatism 'lɪbərəlɪzəm
elitist = an elitist system, government etc is one in which a small group of people have more power and advantages than other people eɪ'li:təst
conservatism= dislike of change and new ideas kən'sɜ:vətɪzəm

READING PASSAGE 2

Tidal Power



Undersea turbines which produce electricity from the tides are **set to** become an important Source of **renewable** energy for Britain. It is still too early to **predict** the **extent** of the impact they may have, but all the signs are that they will **play a significant role** in the future

A
Operating on the same principle as wind **turbines**, the power in sea **turbines** comes from **tidal currents** which turn **blades** similar to ships' **propellers**, but, unlike wind, the tides are

undersea= underwater, submarine, maritime
'ʌndəsi:

turbine= an engine or motor in which the pressure of a liquid or gas moves a special wheel around 'tɜːbaɪn

set to=to start doing something eagerly and with determination set tu:

renewable= renewable energy replaces itself naturally, or is easily replaced because there is a large supply of it rɪ'njuːəbəl

predict= forecast, foresee, guess. prɪ'dɪkt

extent= degree, amount, level, range ɪk'stɛnt

play a role in= to have an effect or influence on something pleɪ ə rəʊl ɪn

operate= work, run, function 'ɒpəreɪt

tidal= relating to the regular rising and falling of the sea. 'taɪdl

current= a continuous movement of water in a river, lake, or sea. 'kʌrənt

blade= the flat wide part of an object that pushes against air or water. bleɪd

propeller= a piece of equipment consisting of two or more blades that spin around, which makes an aircraft or ship move prə'pelə

predictable and the power input is **constant**. The technology raises the **prospect** of Britain becoming **self-sufficient** in **renewable** energy and **drastically** reducing its carbon dioxide **emissions**. If **tide**, wind and wave power are all developed, Britain would be able to close gas, coal and **nuclear power plants** and **export** renewable power to other parts of Europe. Unlike wind power, which Britain originally developed and then **abandoned** for 20 years allowing the Dutch to make it a **major** industry, **undersea turbines** could become a big **export earner** to island nations such as Japan and New Zealand.

B

Tidal sites have already been **identified** that will produce one sixth or more of the UK's power - and at prices **competitive** with modern gas **turbines** and **undercutting** those of the already **ailing nuclear** industry. One **site** alone, the Pentland Firth, between Orkney and **mainland** Scotland, could produce 10% of the country's electricity with **banks of turbines** under the sea, and another at Alderney in the Channel Islands three times the 1,200 megawatts of Britain's largest and newest nuclear **plant**, Sizewell B, in Suffolk. Other sites **identified** include the Bristol Channel and the west coast of Scotland, particularly the channel between Campbeltown and Northern Ireland.

C

Work on designs for the new **turbine blades** and **sites** are well **advanced** at the University of Southampton's **sustainable** energy research group. The first station is **expected** to be installed

constant= regular, stable, steady, even. 'kɒnstənt
prospect= possibility, likelihood, potential. 'prɒspekt
self-sufficient= able to provide all the things you need without help from other people self - sə'fɪʃnt
drastically= extremely, hugely, significantly 'dræstɪklɪ
emission= a gas or other substance that is sent into the air ɪ'mɪʃən
tide= wave, flow, stream, current. taɪd
power plant= a building where electricity is produced to supply a large area paʊə plɑ:nt
export= sell abroad, sell overseas, distribute, ship 'eksɒɪt
abandon= discard, dump, dispose of #keep ə'bændən
major= main, chief, key #minor 'meɪdʒə
earner= a business or activity which makes a profit 'ɜ:nə

identify= recognize, find, detect. aɪ'dentɪfaɪ
competitive= products or prices that are competitive are cheaper than others but still of good quality. kəm'petətɪv
undercut= to sell goods or a service at a lower price than another company. ʌndə'kʌt
ailing= an ailing company, organization, or economy is having a lot of problems and is not successful. 'eɪlɪŋ
site= place, location, spot. saɪt
mainland= the main area of land that forms a country, as compared to islands near it that are also part of that country. 'meɪnlənd
bank= land along the side of a river or lake. bæŋk
plant= a factory or building where an industrial process happens plɑ:nt

work on= to spend time working in order to produce or repair something 'wɜ:k ɒn
advance= improve, develop, enhance, progress əd'vɑ:ns
sustainable= able to continue without causing damage to the environment sə'steɪnəbəl
expect= anticipate, think, believe. ɪk'spekt

off Lynmouth in Devon shortly to test the technology in a **venture jointly funded** by the department of Trade and Industry and the European Union. Abubakr Bahaj, **in charge of** the Southampton research, said: The **prospects** for energy from **tidal currents** are far better than from wind because the flows of water are predictable and **constant**. The technology for **dealing with** the hostile **saline** environment under the sea has been developed in the North Sea oil industry and much is already known about **turbine blade** design, because of wind power and ship **propellers**. There are a few technical difficulties, but I believe in the next five to ten years we will be installing **commercial** marine **turbine** farms.' Southampton has been awarded £215,000 over three years to develop the **turbines** and is working with Marine **Current Turbines**, a **subsidiary** of IT power, on the Lynmouth project. EU research has now **identified** 106 **potential** sites for **tidal** power, 80% **round** the coasts of Britain. The best **sites** are between islands or around heavily **indented** coasts where there are strong **tidal currents**.

D

A marine **turbine blade** needs to be only one third of the size of a wind **generator** to produce three times as much power. The **blades** will be about 20 metres **in diameter**, so around 30 metres of water is required. Unlike wind power, there are unlikely to be environmental **objections**. Fish and other **creatures** are thought unlikely to be **at risk** from the **relatively** slow-turning **blades**. Each **turbine** will be **mounted** on a tower which will connect to the national power supply **grid** via **underwater cables**. The towers will **stick** out of the water and be lit, to warn shipping, and also be designed to be

off= only a short distance away from a place ɒf
venture = a new business activity that involves taking risks. 'ventʃə
jointly = together, cooperatively, mutually. 'dʒɔɪntli
fund = finance, support, back, sponsor. fʌnd
be in charge of something = the position of having control or responsibility for a group of people or an activity bi in tʃɑ:dʒ əv 'sʌmθɪŋ
deal with = cope with, handle, manage. di:l wið
hostile = harsh, adverse, unfavourable 'hɒstail
saline= containing or consisting of salt 'seɪlaɪn
commercial = for profit, trade, business-related. kə'mɜ:ʃəl
subsidiary= a company that is owned or controlled by another larger company. səb'sɪdiəri
potential= possible, likely, latent. pə'tenʃəl
indented= an indented edge or surface has cuts or marks in it ɪn'dentɪd

generator= a machine that produces electricity 'dʒenəreɪtə
diameter= a straight line from one side of a circle to the other side, passing through the centre of the circle, or the length of this line. daɪ'æmɪtə
objection= difficulty, problem, concern. əb'dʒekʃən
creature= animal, living thing, organism. 'kri:tʃə
at risk= in a situation where you may be harmed ət rɪsk
relatively= fairly, quite, rather. 'relatɪvli
mount= install, fit. maʊnt
grid= the network of electricity supply wires that connects power stations and provides electricity to buildings in an area grɪd
cable= rope, chain, line. 'keɪbəl
stick= attach, glue, join, fasten stɪk

lifted out of the water for **maintenance** and to clean **seaweed** from the **blades**.

E

Dr Bahaj has done most work on the Alderney **site**, where there are powerful currents. The single undersea **turbine** farm would produce far more power than needed for the Channel Islands and most would **be fed into** the French **Grid** and be re-**imported** into Britain via the **cable** under the Channel.

F

One technical difficulty is **cavitation**, where low pressure behind a turning **blade** causes air bubbles. These can cause **vibration** and damage the **blades** of the **turbines**. Dr Bahaj said: 'We have to test a number of **blade** types to avoid this happening or at least **make sure** it does not damage the **turbines** or reduce **performance**. Another **slight** concern is **submerged debris** floating into the **blades**. **So far** we do not know how much of a problem it might be. We will have to make the **turbines robust** because the sea is a **hostile** environment, but all the signs that we can do it are good.'

maintenance= the repairs, painting etc that are necessary to keep something in good condition 'meɪntənəns

seaweed= a plant that grows in the sea 'si:wi:d

feed into something= contribute, provide, flow into, deliver. fi:d 'ɪntə 'sʌmθɪŋ

import= a product that is brought from one country into another so that it can be sold there, or the business of doing this #export 'ɪmpɔ:t

cavitation= the forming of gas bubbles in a liquid, caused by changes in pressure kæv.i'teɪ.ʃən

vibration= a continuous slight shaking movement. vaɪ'breɪʃən

make sure (that)= to find out if something is true or to check that something has been done 'meɪk ʃʊə (ðæt)

performance= how well a car or other machine works pə'fɔ:məns

slight= small, minor, unimportant slaɪt

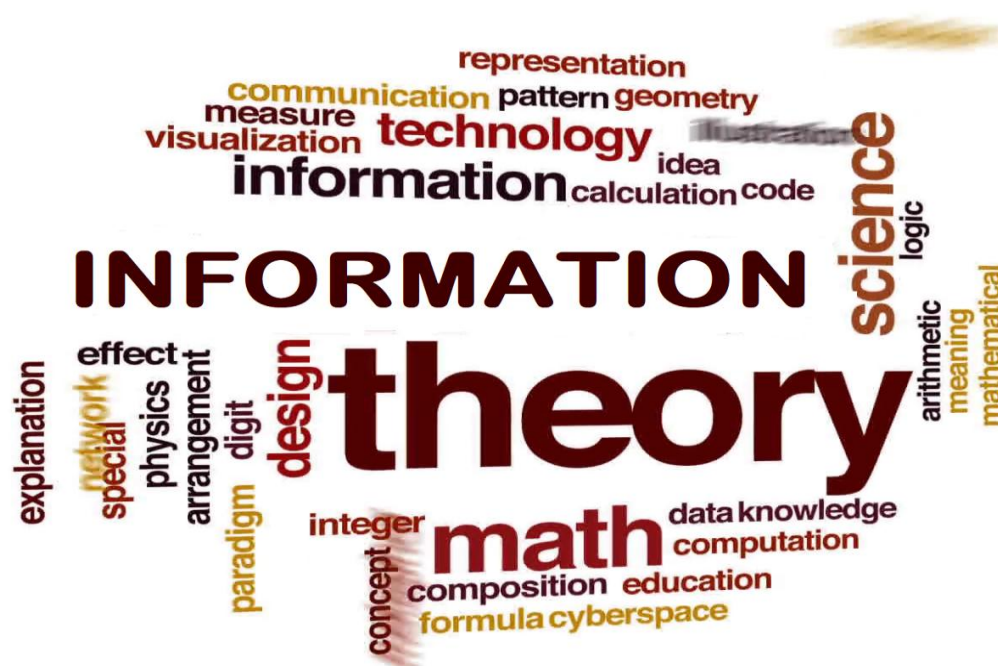
submerged= flooded, underwater, soaked. səb'mɜ:dʒd

debris= rubbish, garbage, trash, waste. 'debri:

so far= until now səʊ fɑ:

robust= strong, healthy, vigorous, tough rə'bʌst

READING PASSAGE 3



Information **theory** lies at the heart of everything

- from DVD players and the **genetic code** of DNA to the physics of the universe at its most **fundamental**. It has been **central** to the development of the science of communication, which **enables** data to be sent electronically and has therefore had a **major impact** on our lives

A

In April 2002 an event **took place** which **demonstrated** one of the many **applications** of information **theory**. The **space probe**, Voyager I,

theory = concept, principle, idea. 'θɪəri
lie at the heart/centre/root of something= to be the most important part of something, especially the main cause of it laɪ ət ðə 'hɑ:t/ sentə/ ru:t əv 'sʌmθɪŋ
genetic code = the arrangement of genes that controls the way a living thing develops. dʒɪ'netɪk kəʊd
fundamental= basic, essential, primary. fʌndə'mentl
central = vital, crucial, fundamental, most important. 'sentrəl
enable = allow, permit, aid, support #prevent ɪ'neɪbəl
major= main, key, chief 'meɪdʒə
impact= influence, effect, impression. 'ɪmpækt

take place= happen, occur, arise. 'teɪk 'pleɪs
demonstrate= show, reveal, display. 'demonstreɪt
application= use, function, purpose. ˌæplɪ'keɪʃən
space probe= spacecraft, spaceship speɪs prəʊb

launched in 1977, had sent back **spectacular** images of Jupiter and Saturn and then **soared** out of **the Solar System** on a one-way mission to the stars. After 25 years of **exposure** to the freezing temperatures of **deep space**, the **probe** was beginning to show its age. **Sensors** and **circuits** were **on the brink of** failing and NASA experts realised that they had to do something or lose contact with their **probe** forever. The **solution** was to get a message to Voyager I to **instruct** it to use **spares** to change the failing parts. With the **probe** 12 billion kilometres from Earth, this was not an easy task. **By means of** a radio dish belonging to NASA's Deep Space Network, the message was sent out into the depths of space. Even travelling at the speed of light, it took over 11 hours to reach its target, far beyond the **orbit** of Pluto. Yet, **incredibly**, the little **probe** managed to hear the **faint** call from its home planet, and successfully made the **switchover**.

B

It was the longest-distance repair job in history, and a **triumph** for the NASA engineers. But it also **highlighted** the **astonishing** power of the techniques developed by American communications engineer Claude Shannon, who had died just a year earlier. Born in 1916 in Petoskey, Michigan, Shannon showed an early talent for maths and for building **gadgets**, and made **breakthroughs** in the **foundations** of

launch= dispatch, send, release *lɔːntʃ*
spectacular= stunning, impressive, fantastic, amazing. *spek'tækjələ*
soar= to fly, especially very high up in the sky, floating on air currents. *sɔː*
the solar system= space, galaxy, universe, cosmos *ðə səʊlə 'sɪstəm*
exposure to something= when someone is in a situation where they are not protected from something dangerous or unpleasant. *ɪk'spəʊʒə tu 'sʌmθɪŋ*
deep space= any region of outer space beyond the system of the earth and moon *'diːp speɪs*
probe= space probe, spacecraft, spaceship *prəʊb*
sensor= a piece of equipment used for discovering the presence of light, heat, movement etc *'sensə*
circuit= trip, journey, route *'sɜːkɪt*
on the brink of something= a situation when you are almost in a new situation, usually a bad one *'ɒn ðə brɪŋk əv 'sʌmθɪŋ*
instruct= order, command, tell. *ɪn'strʌkt*
spare= not being used or not needed at the present time *speə*
by means of sth= using a particular method or system *'baɪ miːnz əv 'sʌmθɪŋ*
orbit= the curved path travelled by an object which is moving around another much larger object such as the earth, the sun etc *'ɔːbɪt*
incredibly= unbelievably, amazing, inconceivably. *ɪn'kredəbli*
manage to do something= to succeed in doing something difficult, especially after trying very hard. *'mænɪdʒ tu duː 'sʌmθɪŋ*
faint= weak, unclear, feeble. *feɪnt*
switchover= a change from using or doing one thing to another *swɪtʃ 'əʊvə*

triumph= victory, achievement, success.

'traɪəmf

highlight= emphasize, stress, underline.

'haɪlaɪt

astonishing= surprising, shocking, amazing. *ə'stɒnɪʃɪŋ*

gadget= device, tool, implement *'gædʒɪt*

breakthrough= advance, invention, innovation, revolution. *'breɪkθruː*

foundation= the establishment of an organization, business, school etc

faʊn'deɪʃən

computer technology when still a student. While at Bell Laboratories, Shannon developed information **theory**, but **shunned** the resulting **acclaim**. In the 1940s, he single-handedly created an entire science of communication which has since **inveigled its way into a host of** applications, from dvds to **satellite** communications to bar codes - any area, **in short**, where data has to be **conveyed** rapidly yet accurately.

C

This all seems **light years** away from the down-to-earth uses Shannon originally had for his work, which began when he was a 22-year-old graduate engineering student at the **prestigious** Massachusetts Institute of Technology in 1939. He **set out** with an **apparently** simple aim: to **pin down** the **precise** meaning of the **concept** of 'information'. The most basic form of information, Shannon **argued**, is whether something is true or false - which can be **captured** in the binary unit, or 'bit', of the form 1 or 0. Having **identified** this **fundamental** unit, Shannon **set about** defining otherwise **vague** ideas about information and how to **transmit** it from place to place. In the process he discovered something surprising: it is always possible to **guarantee** information will **get through** random **interference** - 'noise' - **intact**.

D

Noise usually means unwanted sounds which **interfere** with **genuine** information. Information

shun=avoid, ignore, reject. ʃʌn
acclaim= praise, approval, applause, compliments. ə'kleɪm
inveigle somebody into something= persuade, convince, entice. ɪn'veɪgl 'sʌmbədi 'ɪntə 'sʌmθɪŋ
a host of= a large number of people or things ə hæʊst ɒv
satellite= a machine that has been sent into space and goes around the earth, moon etc, used for radio, television, and other electronic communication 'sætələɪt
in short= briefly, in summary, in a nutshell ɪn ʃɔ:t
convey= pass on, send, transmit. kən'veɪ

light year= the distance that light travels in one year, about 9,460,000,000,000 kilometres, used for measuring distances between stars laɪt 'jiə
prestigious= admired, respected, celebrated, famed, high-status. pre'stɪdʒəs
set out= to start doing something or making plans to do something in order to achieve a particular result set 'aʊt
apparently= obviously, evidently. ə'pærəntli
pin sb/sth down= to understand something clearly or be able to describe it exactly pɪn 'sʌmbədi/ 'sʌmθɪŋ daʊn
precise= exact, correct. accurate. pri'saɪs
concept= idea, notion, thought, belief. 'kɒnsept
argue= say, reason, claim, debate 'ɑ:gju:
capture= describe, portray, depict, denote. 'kæptʃə
identify= discover, find, detect. aɪ'dentɪfaɪ
set about doing sth= start, begin, tackle effort set əbaʊt 'du:ɪŋ 'sʌmθɪŋ
vague= unclear, imprecise, ambiguous veɪg
transmit= communicate, spread, transfer trænz'mɪt
guarantee= assure, ensure. gæ'ren'ti:
get through= to reach a place or person that is difficult to reach 'get θru:
interference= unwanted noise on the radio, television, or on the telephone, or faults in the television picture. ɪntə'fɪərəns
intact= complete, unharmed, integral, undamaged. ɪn'tækt

interfere with= hinder, hamper, restrict, impede. ɪntə'fɪə wɪð
genuine= unaffected, sincere. 'dʒenjuɪn

theory generalises this idea via **theorems** that **capture** the effects of noise with mathematical **precision**. In **particular**, Shannon showed that noise sets a limit on the rate at which information can pass along communication channels while remaining error-free. This rate **depends on** the **relative** strengths of the signal and noise travelling down the communication channel, and on its **capacity** (its '**bandwidth**'). The resulting limit, given in units of bits per second, is the absolute maximum rate of error-free communication given signal strength and noise level. The trick, Shannon showed, is to find ways of **packaging up** - 'coding' - information to **cope with the ravages of** noise, while staying within the information-carrying **capacity** - '**bandwidth**' - of the communication system being used.

E
Over the years scientists have **devised** many such coding methods, and they have proved **crucial** in many technological **feats**. The Voyager **spacecraft** **transmitted** data using codes which added one extra bit for every single bit of information; the result was an error rate of just one bit in 10,000 - and stunningly clear pictures of the planets. Other codes have become part of everyday life - such as the Universal Product Code, or bar code, which uses a simple error-detecting system that **ensures** supermarket **check-out** lasers can read the price even on, say, a **crumpled** bag of **crisps**. As recently as 1993, engineers made a **major breakthrough** by discovering **so-called** turbo codes - which come very close to Shannon's **ultimate** limit for the maximum rate that data can be **transmitted reliably**, and now play a key role in the mobile videophone **revolution**.

generalise= to say that an idea, result etc is related to a larger group 'dʒenərəlaɪz
theorem= theory, formula, statement 'θiərəm
capture= acquire, gain, attain. 'kæptʃə
precision= accuracy, correctness, exactness prɪ'sɪʒən
in particular= especially, specially, in general ɪn pə'tɪkjələ
depend on= rely on, count on, hinge on dɪ'pend ɒn
relative= having a particular quality when compared with something else 'relatɪv
capacity= size, volume, amount. kə'pæsəti
bandwidth= the amount of information that can be carried through a telephone wire, computer connection etc at one time 'bændwɪðθ
package up= to put food or other goods into a bag, box etc ready to be sold or sent 'pækɪdʒ ʌp
cope with= deal with, tackle, handle. kəʊp wɪð
the ravages of something= the damage caused by something. ðə 'rævɪdʒɪz ɒv 'sʌmθɪŋ

devise= invent, develop, create. dɪ'vaɪz
crucial= vital, central, most important. 'kruːʃəl
feat= achievement, accomplishment, deed. fi:t
spacecraft= space probe, probe, spaceship 'speɪs-kraːft
ensure= make sure, guarantee, make certain. ɪn'ʃʊə
check out= to make sure that something is actually true, correct, or acceptable tʃek 'aʊt
crumpled= crushed into a smaller bent shape 'krʌmpld
crisp= a very thin flat round piece of potato that is cooked in oil and eaten cold. krɪsp
so-called= used to show that something or someone is usually called a particular name səʊ - 'kɔːld
ultimate= final, last, eventual. 'ʌltəmət
revolution= change, development, innovation. revə'luːʃən

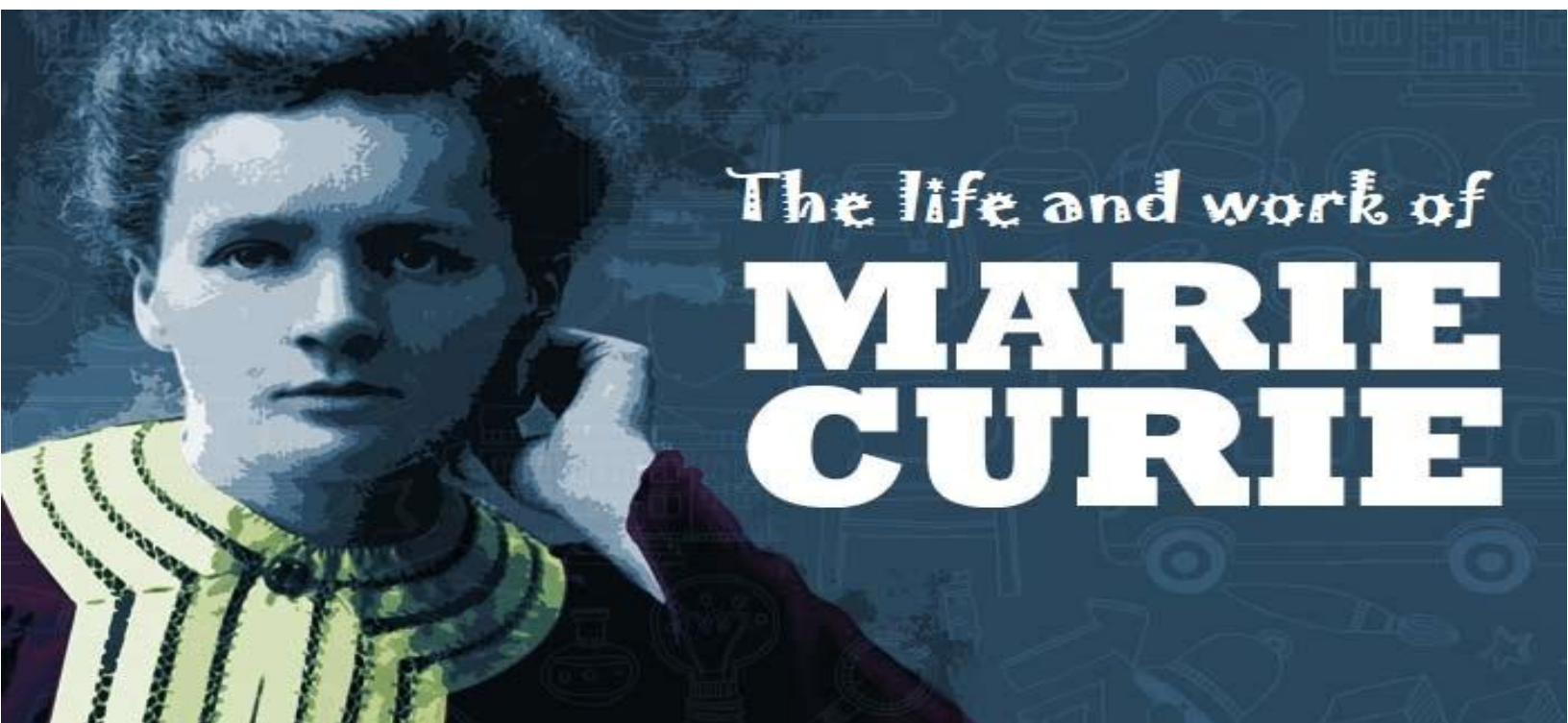
F

Shannon also **laid the foundations** of more **efficient** ways of storing information, by stripping out **superfluous** ('redundant') bits from data which **contributed** little real information. As mobile phone text messages like 'I CN C U' show, it is often possible to **leave out** a lot of data without losing much meaning. As with error correction, however, there's a limit beyond which messages become too **ambiguous**. Shannon showed how to calculate this limit, **opening the way** to the design of **compression** methods that **cram maximum information into the minimum space**.

lay the foundations/groundwork/base= to provide the conditions that will make it possible for something to happen or be successful 'leɪ ðə faʊn'deɪʃnz/ 'graʊndwɜ:k/ beɪs
efficient= if someone or something is efficient, they work well without wasting time, money, or energy ɪ'fɪʃənt
strip= remove, rid, take away. strɪp
superfluous= surplus, unnecessary, excessive, redundant. su:'pɜ:fluəs
redundant= superfluous. rɪ'dʌndənt
contribute= add, give, provide. kən'trɪbjʊ:t
leave out= exclude, ignore, omit. 'li:v 'aʊt
ambiguous= unclear, uncertain, confusing. æm'bigjuəs
open the door/way to sth= to make an opportunity for something to happen 'əʊpən ðə 'dɔ: 'weɪ tu 'sʌmθɪŋ
compression= the act of making something smaller or shorter so that it will fit into a particular space or time kəm'preʃən
cram something into/onto etc something = to force something into a small space. kræm 'sʌmθɪŋ 'ɪntə/ 'ɒntu et 'setrə 'sʌmθɪŋ

TEST 4

READING PASSAGE 1



Marie Curie is **probably** the most famous woman scientist who has ever lived. Born Maria Sklodowska in Poland in 1867, she is famous for her work on **radioactivity**, and was twice a winner of the Nobel Prize. With her husband, Pierre Curie, and Henri Becquerel, she was **awarded** the 1903 Nobel Prize for Physics, and was then **sole** winner of the 1911 Nobel Prize for Chemistry. She was the first woman to win a Nobel Prize.

From childhood, Marie was **remarkable** for her **prodigious** memory, and at the age of 16 won a gold **medal on completion of** her **secondary education**.

probably= perhaps, maybe, possibly

'prɒbəbli

radioactivity : the sending out of radiation (=a form of energy) when the nucleus (=central part) of an atom has broken apart *reɪdɪəʊæk'tɪvɪti*

award= prize, reward, gift. *ə'wɔ:d*

sole= only, single, individual. *səʊl*

remarkable= outstanding, noteworthy, extraordinary *rɪ'mɑ:kəbəl*

prodigious= extraordinary, phenomenal, unusual, remarkable, impressive. *prə'dɪdʒəs*

on completion of= the act of finishing something *'ɒn kəm'pli:ʃŋ ɒv*

secondary education= the education, teaching etc of children between the ages of 11 and 16 or 18 *'sekəndri edʒu'keɪʃn*

Because her father lost his savings through bad investment, she then had to take work as a teacher.

From her **earnings** she was able to **finance** her sister Bronia's medical studies in Paris, **on the understanding** that Bronia would, **in turn**, later help her to get an education.

In 1891 this promise was **fulfilled** and Marie went to Paris and began to study at the Sorbonne (the University of Paris). She often worked far into the night and **lived on** little more than **bread and butter** and tea. She came first in the **examination** in the physical sciences in 1893, and in 1894 was placed second in the **examination** in mathematical sciences. It was not until the spring of that year that she was introduced to Pierre Curie.

Their marriage in 1895 **marked** the start of a **partnership** that was soon to achieve results of world significance. Following Henri Becquerel's discovery in 1896 of a new **phenomenon**, which Marie later called '**radioactivity**', Marie Curie decided to **find out** if the **radioactivity** discovered in uranium was to be found in other elements. She discovered that this was true for thorium.

Turning her attention to **minerals**, she found her interest **drawn** to **pitchblende**, a **mineral** whose **radioactivity**, **superior** to that of **pure** uranium, could be explained only by the presence in the **ore** of small

earnings= salary, wage, income. 'ɜ:nɪŋz
finance= support, fund, back, sponsor.
'faɪnæns

on the understanding that= if you agree to something on the understanding that something else will be done, you agree to it, believing that it will be done 'ɒn ði ʌndə'stændɪŋ ðæt

in turn= one after the other in an agreed order ɪn tɜ:n

fulfill (a promise) = accomplish, complete, finish, carry out. fʊl'fɪl (ə 'prɒmɪs)
far into the night= extending until a late hour 'fɑ:r 'ɪntə ðə naɪt

live on sth=to have a particular amount of money to buy food and other necessary things 'laɪv 'ɒn 'sʌmθɪŋ
bread and butter (work)= work that is not very exciting but provides you with most of the money that you need in order to live bred ənd 'bʌtə ('wɜ:k)

examination= exam, test, assessment.
ɪg,zæmə'neɪʃən

mark= to be a sign of an important change or an important stage in the development of something mɑ:k

partnership= collaboration, affiliation, companionship. 'pɑ:tənəʃɪp

phenomenon= occurrence, experience, event. fɪ'nomənən

find out= realize, learn, discover. faɪnd 'aʊt

mineral= a substance that is formed naturally in the earth, such as coal, salt, stone, or gold. minerals can be dug out of the ground and used. 'mɪnərəl

draw sb to sth= to attract someone or make them want to do something drɔ: 'sʌmbədi tu 'sʌmθɪŋ

pitchblende= a dark shiny substance dug from the earth, from which uranium and radium are obtained 'pɪtʃblend

superior = better quality, greater. advanced, enhanced #inferior. su:'prɪəriə

pure = unmixed, real, genuine. pjʊə

ore = rock or earth from which metal can be obtained ɔ:

quantities of an unknown **substance** of very high activity. Pierre Curie joined her in the work that she had **undertaken** to **resolve** this problem, and that led to the discovery of the new elements, polonium and radium. While Pierre Curie **devoted himself chiefly** to the physical study of the new radiations, Marie Curie **struggled** to **obtain** pure radium in the **metallic** state. This was achieved with the help of the chemist André-Louis Debierne, one of Pierre Curie's pupils. Based on the results of this research, Marie Curie received her **Doctorate** of Science, and in 13 Marie and Pierre shared with Becquerel the Nobel Prize for Physics for the discovery of **radioactivity**.

The births of Marie's two daughters, Irène and Eve, in 1897 and 1904 failed to **interrupt** her scientific work. She was **appointed** lecturer in physics at the École Normale Supérieure for girls in Sèvres, France (1900), and introduced a method of teaching based on **experimental demonstrations**. In December 1904 she was **appointed** **chief** assistant in the laboratory directed by Pierre Curie.

The sudden death of her husband in 1906 was a **bitter** blow to Marie Curie, but was also a **turning point** in her career: **henceforth** she was to **devote** all her energy to completing alone the scientific work that they had **undertaken**. On May 13, 1906, she was **appointed** to the **professorship** that had been left **vacant** on her husband's death, becoming the first woman to teach at the Sorbonne. In 1911 she was

substance = element, material, ingredient, element. 'sʌbstəns
undertake to do sth = to promise or agree to do something ʌndə'teɪk tu du: 'sʌmθɪŋ
resolve = to find a satisfactory way of dealing with a problem or difficulty = solve, work out. rɪ'zɒlv
devote your time/ energy/ attention/ yourself etc to something = dedicate = to use all or most of your time, effort etc in order to do something or help someone. dɪ'vəʊt jə 'taɪm/ 'enədʒi/ ə'tenʃn/ jə: 'self et' setrə tu 'sʌmθɪŋ
chiefly = primarily, mainly, mostly. 'tʃi:fli
radiation = a form of energy that comes especially from nuclear reactions, which is very harmful to living things. reɪdɪ'eɪʃn
struggle to do sth = to try extremely hard to achieve something, even though it is very difficult 'strʌgl tu du: 'sʌmθɪŋ
obtain = get, gain, find, acquire. əb'teɪn
metallic = made of metal or containing metal mə'tælɪk
doctorate = a university degree of the highest level 'dɒktərət

interrupt = disrupt, suspend, stop. ɪntə'rʌpt
appoint = assign, select, choose. ə'pɔɪnt
experimental = used for, relating to, or resulting from experiments ɪk'sperə'menti
demonstration = an act of explaining and showing how to do something or how something works demən'streɪʃn
chief = highest in rank tʃi:f

bitter = making you feel very unhappy and upset 'bɪtə
blow = an action or event that causes difficulty or sadness for someone bləʊ
turning point = the time when an important change starts, especially one that improves the situation 'tɜ:nɪŋ poɪnt
henceforth = from this time on hens'fɔ:θ
professorship = the job or position of a university or college professor prə'fesəʃɪp
vacant = available, untaken, unused, free. 'veɪkənt

awarded the Nobel Prize for Chemistry for the **isolation** of a **pure** form of radium.

During World War I, Marie Curie, with the help of her daughter Irène, **devoted** herself to the development of the use of X-**radiography**, including the **mobile** units which came to be known as 'Little Curies', used for the **treatment** of **wounded** soldiers. In 1918 the Radium Institute, whose staff Irène had joined, began to **operate in earnest**, and became a centre for nuclear physics and chemistry. Marie Curie, now at the highest point of her **fame** and, from 1922, a member of the Academy of Medicine, researched the chemistry of **radioactive substances** and their medical **applications**.

In 1921, **accompanied** by her two daughters, Marie Curie made a **triumphant** journey to the United States to raise **funds** for research on radium. Women there presented her with a gram of radium for her **campaign**. Marie also gave lectures in Belgium, Brazil, Spain and Czechoslovakia and, in addition, had the satisfaction of seeing the development of the Curie **Foundation** in Paris, and the **inauguration** in 1932 in Warsaw of the Radium Institute, where her sister Bronia became director.

One of Marie Curie's **outstanding** achievements was to have understood the need to **accumulate intense** **radioactive** sources, not only to treat illness but also to maintain an **abundant** supply for research. The

isolation= separation, segregation
#inclusion aɪsəˈleɪʃən

radiography= the taking of x-ray photographs of the inside of people's bodies for medical purposes reɪdɪˈɒɡrəfi

mobile= moveable, portable, active. ˈməʊbaɪl

treatment= cure, healing, medicine, therapy. ˈtriːtmənt

wounded= injured, hurt, maimed. ˈwuːndɪd

operate = run, work, conduct, carry out. ˈɒpəreɪt

in earnest= if something starts happening in earnest, it begins properly - used when it was happening in a small or informal way before ɪnˈɜːnɪst

fame= the state of being known about by a lot of people because of your achievements feɪm

application= practical purpose for which a machine, idea etc can be used, or a situation when this is used æplɪˈkeɪʃən

accompany= attend, go with, go along with. əˈkʌmpəni

triumphant= successful, winning, victorious. traɪˈʌmfənt

funds= money that an organization needs or has fʌndz

campaign= a series of actions intended to achieve a particular result relating to politics or business, or a social improvement kæmˈpeɪn

foundation= an organization that gives or collects money to be used for special purposes, especially for charity or for medical research faʊnˈdeɪʃən

inauguration= the act of officially putting someone into an important position, or the ceremony at which this is done. ɪˌnɔːɡjʊˈreɪʃən

outstanding= excellent, great, remarkable aʊtˈstændɪŋ

accumulate= gather, collect, amass, pile up. əˈkjuːmjəleɪt

intense=having a very strong effect or felt very strongly. ɪnˈtens

abundant = plentiful, rich, ample #scarce. əˈbʌndənt

existence in Paris at the Radium Institute of a stock of 1.5 grams of radium made a **decisive** contribution to the success of the experiments **undertaken** in the years around 1930. This work **prepared the way for** the discovery of the **neutron** by Sir James Chadwick and, **above all**, for the discovery in 1934 by Irène and Frédéric Joliot-Curie of **artificial radioactivity**. A few months after this discovery, Marie Curie died as a result of **leukaemia** caused by **exposure** to **radiation**. She had often carried **test tubes** containing **radioactive isotopes** in her pocket, remarking on the pretty blue-green light they **gave off**.

Her contribution to physics had been **immense**, not only in her own work, the importance of which had been **demonstrated** by her two Nobel Prizes, but because of her influence on **subsequent** generations of nuclear physicists and chemists.

decisive= key, critical, significant, vital. dɪ'saɪsɪv
prepare the way/ground for sb/sth= to make it possible for something to be achieved, or for someone to succeed in doing something prɪ'peə ðə 'weɪ graʊnd fə 'sʌmbədi/ 'sʌmθɪŋ

neutron= a part of an atom that has no electrical charge 'nju:trɒn

above all= more than everything else əbʌv ɔ:l

artificial= man-made, non-natural, synthetic
 #real, natural ɑ:tɪ'fɪʃəl

leukaemia = a type of cancer of the blood, that causes weakness and sometimes death. lu:'ki:miə

exposure to something = when someone is in a situation where they are not protected from something dangerous or unpleasant. ɪk'spəʊʒə tu 'sʌmθɪŋ

test tube= a small glass container that is shaped like a tube and is used in chemistry 'test 'tju:b

isotopes = one of the possible different forms of an atom of a particular element (=simple chemical substance) 'aɪsə'təʊps

give off= to produce a smell, light, heat, a sound etc gɪv ɒf

immense = great, huge, enormous.

ɪ'mens

demonstrate= show, reveal, display

'demonstreɪt

subsequent= following, consequent, later

'sʌbsəkwənt

READING PASSAGE 2



A

A sense of self develops in young children

by **degrees**. The process can usefully be thought of in **terms of** the **gradual emergence** of two **somewhat separate** features: the self as a subject, and the self as an object. William James introduced the **distinction** in 1892, and **contemporaries** of his, such as Charles Cooley,

somebody's sense of self = someone's idea that they are a separate person, different from other people 'səm,bɔːdi sɛns əv self

degree= level, scale, extent. dɪ'ɡriː

in terms of= if you explain or describe something in terms of a particular fact or event, you are explaining or describing it only in relation to that fact or event ɪn tɜːmz ɒv

gradual= slow, steady #rapid 'grædʒuəl

emergence= appearance, occurrence, arrival. ɪ'mɜːdʒəns

somewhat = rather, slightly, to some extent 'sʌmwɒt

separate= different, distinct, discrete. 'seperət

feature= a part of something that you notice because it seems important, interesting, or typical 'fi:tʃə

distinction=difference, separation, discrepancy

#similarity dɪ'sɪŋkjən

contemporary= someone who lived or was in a particular place at the same time as someone else. kən'tempərəri

added to the **developing debate**. Ever since then **psychologists** have continued **building on** the **theory**.

B

According to James, a child's first step on the road to self-**understanding** can be seen as the **recognition** that he or she exists. This is an **aspect** of the self that he **labelled** 'self-as-subject', and he gave it **various elements**. These included an **awareness of** one's own agency (i.e. One's power to act), and an **awareness of** one's **distinctiveness** from other people. These **features** gradually **emerge** as **infants** explore their world and **interact** with **caregivers**. Cooley (1902) suggested that a **sense** of the self-as-subject was **primarily concerned with** being able to exercise power. He **proposed** that the earliest examples of this are an **infant's attempts** to control **physical** objects, such as toys or his or her own **limbs**. This is followed by **attempts** to affect the behaviour of other people. For example, **infants** learn that when they cry or smile someone **responds to** them.

C

Another powerful source of information for **infants** about the effects they can have on the world around them is provided when others **mimic** them. Many parents spend a lot of time, **particularly** in the early months, copying their

developing= growing or changing *dɪ'veləpɪŋ*
debate= argument, discussion, dispute. *dɪ'beɪt*
ever since=all the time since *'evə sɪns*
psychologist= someone who is trained in psychology (the study of the mind and how it influences people's behaviour) *sai'kɒlədʒɪst*
build on sth=to use your achievements as a base for further development *'bɪld 'ɒn 'sʌmθɪŋ*
theory= concept, idea, principle *'θiəri*

understanding= grasp, knowledge, perception *ʌndə'stændɪŋ*

recognition= the act of realizing and accepting that something is true or important *rekəg'nɪʃən*

aspect= feature, part, trait. *'æspekt*

label= mark, consider, describe. *'leɪbəl*

various = many, a range of, numerous. *'veəriəs*

element= part, component, factor. *'eləmənt*

awareness of= the ability to notice something using your senses *ə'veəneəs ɒv*

distinctiveness= something that is distinctive is easy to recognize because it is different from other things *dɪ'stɪŋktɪvnɪs*

emerge = arise, develop, appear. *ɪ'mɜːdʒ*

infant = a baby or very young child. *'ɪnfənt*

interact= if people interact with each other, they talk to each other, work together etc *ɪntər'ækt*

caregiver = someone who takes care of a child or sick person. *'keə,gɪvə*

primarily= mainly, chiefly, mostly. *'praɪməri*

concerned with= involved in something or affected by it *kən'sɜːnd wɪð*

propose = suggest, offer, recommend. *prə'pəʊz*

attempt = try, effort, endeavor. *ə'tempt*

physical= relating to real objects that you can touch, see, or feel *'fɪzɪkəl*

limb= an arm or leg *lɪm*

respond to= react, counter, take action *rɪ'spɒnd tuː*

mimic= imitate, copy, mirror, simulate. *'mɪmɪk*

particularly= especially, specifically, exceptionally. *pə'tɪkjʊləli*

infant's vocalizations and **expressions**. In addition, young children enjoy looking in mirrors, where the movements they can see are **dependent upon** their own movements. **This is not to say** that infants **recognize** the **reflection** as their own image (a later development). However, Lewis and Brooks-Gunn (1979) suggest that **infants'** developing **understanding** that the movements they see in the mirror are **contingent on** their own, leads to a growing awareness that they are **distinct** from other people. This is because they, and only they, can change the **reflection** in the mirror.

D

This **understanding** that children **gain** of themselves as active **agents** continues to develop in their **attempts** to **co-operate** with others in play. Dunn (1988) **points out** that it is in such **day-to-day** relationships and **interactions** that the child's **understanding** of his- or herself **emerges**. **Empirical** investigations of the self-as-subject in young children are, however, rather **scarce** because of difficulties of communication: even if young **infants** can **reflect on** their experience, they certainly cannot **express** this **aspect** of the self directly.

E

Once children have **acquired** a certain level of **self-awareness**, they begin to place themselves in a whole **series** of categories, which together play such an important part in **defining** them **uniquely** as 'themselves'. This second step in

vocalization= a word or sound that is produced by the voice *vəʊkəlaɪ'zeɪʃən*
expression= a look on someone's face that shows what they are thinking or feeling *ɪk'spreʃən*
be dependent on/upon sth= to be directly affected or decided by something else *bi dɪ'pendənt 'ɒn/ ə'pɒn 'sʌmθɪŋ*
this is not to say= used to make sure the person you are talking to does not think something that is not true *ðɪs ɪz nɒt tu 'seɪ*
recognize= know, spot, identify. *'rekəɡnaɪz*
reflection= an image that you can see in a mirror, glass, or water. *rɪ'flekʃən*
contingent on/upon something= depending on something that may happen in the future. *kən'tɪndʒənt 'ɒn/ ə'pɒn 'sʌmθɪŋ*
distinct= different, dissimilar, discrete. *dɪ'stɪŋkt*

gain= get, achieve, acquire, obtain *ɡeɪn*
agent= someone or something that affects or changes a situation *'eɪdʒənt*
co-operate= to work with someone else to achieve something that you both want *kəʊ'ɒpəreɪt*
point out= to tell someone something that they did not already know or had not thought about *pɔɪnt 'aʊt*
day-to-day= day-to-day jobs or activities are ones that you do every day as a normal part of your life, your job etc *'deɪ-tə'deɪ*
interaction= communication, contact, interface *ɪntər'ækʃən*
empirical= based on scientific testing or practical experience, not on ideas= experimental, observed, practical *ɪm'pɪrɪkəl*
scarce= rare, uncommon, unusual *skeəs*
reflect on= to think carefully about something, or to say something that you have been thinking about *rɪ'flekt ɒn*
express= say, state, utter, convey *ɪk'spres*

acquire= obtain, get, attain. *ə'kwəɪə*
self-awareness= knowledge and understanding of yourself *self.ə'weə.nəs*
series= chain, string, sequence *'sɪəri:z*
define= describe, express, state *dɪ'faɪn*
unique= distinctive, only one of it's kind, sole, single, exclusive. *ju:'ni:k*

the development of a full **sense** of self is what James called the 'self-as-object'. This has been seen by many to be the **aspect** of the self which is most **influenced** by social **elements**, since it is **made up** of social roles (such as student, brother, colleague) and **characteristics** which **derive their meaning from comparison** or **interaction** with other people (such as **trustworthiness**, shyness, sporting ability).

F

Cooley and other researchers suggested a close **connection** between a person's own **understanding** of their **identity** and other people's **understanding** of it. Cooley believed that people **build up** their **sense** of **identity** from the reactions of others to them, and from the view they believe others have of them. He called the self-as-object the 'looking-glass self', since people come to see themselves as they are reflected in others. Mead (1934) went even further, and saw the self and the social world as **inextricably bound together**: 'The self is essentially a social structure, and it **arises** in social experience. It is impossible to **conceive** of a self **arising** outside of social experience.'

G

Lewis and Brooks-Gunn **argued** that an important developmental **milestone** is reached when children become able to **recognize** themselves **visually** without the support of seeing **contingent** movement. This **recognition** **occurs** around their second birthday. In one **experiment**, Lewis and Brooks-Gunn (1979)

influence= affect, motivate, inspire. 'ɪnfluəns
be made up of= to combine together to form something bi 'meɪd 'ʌp ɒv
characteristic= trait, feature, quality. kærəktə'rɪstɪk
derive something from something= to get something, especially an advantage or a pleasant feeling, from something. dɪ'raɪv 'sʌmθɪŋ frəm 'sʌmθɪŋ
trustworthiness= the quality or fact of being trustworthy (= able to be trusted) 'trʌstwɜːðɪnəs

connection= link, relationship, association kə'nekʃən
identity= the qualities and attitudes that a person or group of people have, that make them different from other people. aɪ'dentəti
build up= develop, increase, accumulate. 'bɪld 'ʌp
inextricably= if two or more things are inextricably linked etc, they are very closely related and affect each other ɪnɪk'strɪkəbli
bind somebody/something together= unite= to form a strong emotional or economic connection between two people, countries etc. baɪnd 'sʌmbədi/ 'sʌmθɪŋ tə'geðə
arise= happen, occur, take place, start. ə'raɪz
conceive of (doing) something = to imagine a particular situation or to think about something in a particular way. kən'siːv əv ('duːɪŋ) 'sʌmθɪŋ

argue= say, reason, debate, dispute 'ɑːɡjuː
milestone= a very important event in the development of something. = landmark, breakthrough 'maɪlstəʊn
visually= in a way that involves the eyes 'vɪʒuəli
contingent= dependent= depending on something that may happen in the future. kən'tɪndʒənt
occur= happen, arise, ensue ə'kɜː
experiment= test, research, trial. ɪk'sperɪmənt

dabbed some red powder on the noses of children who were playing in front of a mirror, and then **observed** how often they touched their noses. The **psychologists reasoned** that if the children knew what they usually looked like, they would be surprised by the unusual red mark and would start touching it. On the other hand, they found that children of 15 to 18 months are generally not able to **recognize** themselves unless other **cues** such as movement are present.

H

Finally, perhaps the most **graphic expressions** of **self-awareness in general** can be seen in the **displays of rage** which are most common from 18 months to 3 years of age. In a **longitudinal study** of groups of three or four children, Bronson (1975) found that the **intensity** of the **frustration** and **anger** in their disagreements increased **sharply** between the ages of 1 and 2 years. Often, the children's disagreements involved a **struggle** over a toy that none of them had played with before or after the **tug-of-war**: the children seemed to be **disputing ownership** rather than wanting to play with it. Although it may be less **marked** in other societies, the link between the sense of 'self' and of '**ownership**' is a **notable feature** of **childhood** in Western societies.

dab sth on/onto etc sth= to put a substance onto something with quick light movements of your hand dæb 'sʌmθɪŋ 'ɒn 'ɒntu et'setə 'sʌmθɪŋ
observe= watch, view, monitor, examine əb'zɜ:v
reason (that)= to form a particular judgment about a situation after carefully considering the facts 'ri:zən (ðæt)

cue= signal, sign, hint, clue, key kju:

graphic= visual, pictorial, illustrative 'græfɪk
in general= usually, in most cases, overall. ɪn 'dʒenrəl

display= show, exhibit, presentation. dɪ'spleɪ

rage= anger, fury, wrath reɪdʒ

longitudinal study/survey/research etc= relating to the development of something over a period of time. lɒŋgɪ'tju:dɪnəl 'stʌdi/ sə'veɪ/ rɪ'sɜ:tʃ et'setə

intensity= the quality of being serious and having very strong feelings or opinions ɪn'tensəti

frustration= the feeling of being annoyed, upset, or impatient, because you cannot control or change a situation, or achieve something.

fɹʌ'streɪʃən

sharply= suddenly and by a large amount 'ʃɑ:pli

struggle= fight, battle, scrap 'strʌɡəl

tug-of-war = a situation in which two people or groups try very hard to get or keep the same thing tʌɡ əv 'wɔ:r

dispute= to try to get control of something or win something dɪ'spju:t

ownership= possession, right, title 'əʊnəʃɪp

marked= clear, obvious, noticeable. mɑ:kt

notable= important, significant, prominent, outstanding. 'nəʊtəbəl

childhood= the period of time when you are a child 'tʃaɪldhʊd

READING PASSAGE 3

The Development of Museums



The **conviction** that historical **relics** provide

infallible testimony about the past is **rooted** in the nineteenth and early twentieth centuries, when science was regarded as **objective** and **value free**. As one writer **observes**: 'Although it is now **evident** that **artefacts** are as easily **altered** as **chronicles**, public **faith** in their **veracity endures**: a **tangible relic** seems

conviction= strong belief, opinion or view

kən'vɪkʃən

relic= an old object or custom that reminds people of the past or that has lived on from a past time.

'reɪlɪk

infallible= always right, perfect, reliable, dependable

ɪn'fæləbəl

testimony= indication, proof, evidence.

be rooted in sth= to have developed from something and be strongly influenced by it

bi'ru:tɪd ɪn 'sʌmθɪŋ

objective= based on facts, or making a decision that is based on facts rather than on your feelings or beliefs.

əb'dʒektɪv

value-free= making or having no value judgments

'vælju: - fi:

observe= to say or write what you have noticed about a situation

əb'zɜ:v

evident= clear, obvious, apparent.

artefact= an object such as a tool, weapon etc that was made in the past and is historically important.

'ɑ:təfækt

alter= change, modify, adjust, vary #maintain

'ɔ:lte

chronicle= story, record, narrative.

'kronɪkəl

faith= belief, conviction, trust

feɪθ

veracity= truth, accuracy, reliability.

və'ræsəti

endure= to remain alive or continue to exist for a long time.

ɪn'dʒʊə

tangible= if something is tangible, you can touch or feel it.

'tændʒəbəl

ipso facto real.' Such **conviction** was, until recently, reflected in museum **displays**.

Museums used to look - and some still do - much like storage rooms of objects packed together in **showcases**: good for **scholars** who wanted to study the **subtle** differences in design, but not for the ordinary visitor, to whom it all looked alike.

Similarly, the information **accompanying** the objects often **made** little **sense** to the **lay** visitor.

The **content** and format of explanations dated back to a time when the museum was the **exclusive domain** of the scientific researcher.

B

Recently, however, attitudes towards history and the way it should be presented have **altered**. The key word in **heritage display** is now 'experience', the more exciting the better and, if possible, involving all the senses. Good examples of this **approach** in the UK are the Jorvik Centre in York; the National Museum of Photography, Film and Television in Bradford; and the Imperial War Museum in London. In the US the trend **emerged** much earlier: Williamsburg has been a **prototype** for many **heritage** developments in other parts of the world. No one can **predict** where the process will end. On **so-called heritage** sites the

ipso facto= used to show that something is known from or proved by the facts. *ipsəu 'fæktəu*

display= show, exhibition, presentation *di'spleɪ*

showcase= a glass box containing objects for people to look at in a shop, at an art show etc *'ʃəʊkeɪs*

scholar= researcher, academic, specialist *'skɒlə*

subtle= not easy to notice or understand unless you pay careful attention ≠ obvious *'sʌtl*

accompany= go together with. *ə'kʌmpəni*

make sense= to have a clear meaning and be easy to understand *'meɪk sens*

lay= not trained or not knowing much about a particular profession or subject *leɪ*

content= the things that are written in a letter, book etc *'kɒntent*

exclusive= available or belonging only to particular people, and not shared. *ɪk'skluːsɪv*

domain= area, field, sphere, realm. *də'meɪn*

heritage= the traditional beliefs, values, customs etc of a family, country, or society. *'herətɪdʒ*

approach= method, tactic, style *ə'prəʊtʃ*

emerge= appear, begin, come out. *ɪ'mɜːdʒ*

prototype= example, model, sample. *'prəʊtətaɪp*

so-called= used to describe someone or something that has been given a name that you think is wrong *səʊ 'kɔːld*

re-enactment of historical events is increasingly popular, and computers will soon provide **virtual** reality experiences, which will present visitors with a **vivid** image of the **period** of their choice, in which they themselves can act as if part of the historical environment. Such developments have been **criticised** as an **intolerable vulgarisation**, but the success of many historical **theme parks** and similar locations suggests that the **majority** of the public does not **share this opinion**.

C

In a related development, the **sharp distinction** between museum and heritage sites on the one hand, and **theme parks** on the other, is gradually **evaporating**. They already borrow ideas and **concepts** from one another. For example, museums have **adopted** story lines for exhibitions, sites have accepted 'theming' as a **relevant** tool, and **theme parks** are moving towards more **authenticity** and research-based presentations. In zoos, animals are no longer kept in cages, but in great spaces, either in the **open air** or in **enormous greenhouses**, such as the jungle and desert environments in

re-enactment= an activity that repeats the actions of a past event, especially as an entertainment
ri:ɪ' nækt.mənt
virtual= made, done, seen etc on the internet or on a computer, rather than in the real world 'vɜ:tʃuəl
vivid= vivid memories, dreams, descriptions etc are so clear that they seem real. 'vɪvɪd
period= a particular time in someone's life or in history 'pɪəriəd
criticise= complain, condemn, disapprove #praise 'krɪtɪsaɪz
intolerable= too difficult, bad, annoying etc for you to accept or deal with. ɪn'tolərəbəl
vulgarisation= the process of spoiling something by changing it so that it is more ordinary than before and not of such a high standard
vʌlgəraɪ'zeɪʃən
theme park= a type of park where you can have fun riding on big machines such as a roller coaster, and where the whole park is based on one subject such as water or space travel θi:m pɑ:k
majority= most of the people or things in a group #minority mə'ɔːrəti
share sb's view/concern/belief etc= to have the same opinion, quality, or experience as someone else 'ʃeə 'səm,bɑ:di vju:/ kən'sɜ:n/ br'i:lɪf et'setə
sharp= sharp differences are very big and very noticeable ʃɑ:p
distinction= difference, discrepancy #similarity dɪ'stɪŋkʃən
evaporate= if a feeling evaporates, it slowly disappears = vanish, fade, dissolve #solidify ɪ'væpəreɪt
concept= idea, view, belief 'kɒnsept
adopt= approve, accept, embrace.
story line= plot= the main set of related events in a story ə'dɒpt
theme= a particular style
relevant= directly relating to the subject or problem being discussed or considered #irrelevant θi:m
authenticity= the quality of being real or true ɔ:θen'tɪsəti
-based= used to describe the basic feature or part of something beɪst
open air= outdoor= happening or existing outdoors 'əʊpən 'eə
enormous= huge, vast, massive #tiny ɪ'no:məs
greenhouse= a glass building used for growing plants that need warmth, light, and protection 'ɡri:nhaʊs

Burgers' Zoo in Holland. This particular **trend** is **regarded** as one of the major developments in the presentation of natural history in the twentieth century.

D

Theme parks are **undergoing** other changes, too, as they try to present more **serious** social and cultural issues, and **move away** from fantasy. This development is a response to **market forces** and, although museums and **heritage** sites have a special, rather distinct, role to **fulfill**, they are also **operating** in a very competitive environment, where visitors make choices on how and where to spend their free time. **Heritage** and museum **experts** do not have to invent stories and **recreate** historical environments to attract their visitors: their **assets** are already in place. However, **exhibits** must be both based on **artefacts** and facts as we know them, and attractively presented. Those who are professionally **engaged in** the art of **interpreting** history are thus in a difficult position, as they must **steer a narrow course** between the demands of 'evidence' and 'attractiveness', especially given the increasing need in the **heritage** industry for income-generating activities.

E

It could be claimed that in order to make everything in **heritage** more 'real', historical

trend= a general tendency in the way a situation is changing or developing trend
regard= consider, think, see rɪ'gɑ:d

undergo= experience, endure, go through.

ʌndə'geʊ

serious= important, significant, crucial 'sɪəriəs

fulfill= satisfy. fʊl'fɪl

move away= to start talking or writing about a different subject mu:v ə'wei

market forces= the way that the behaviour of buyers and sellers affects the levels of prices and wages, without any influence from the government 'mɑ:kɪt 'fɔ:sɪz

fulfill= accomplish, achieve, satisfy fʊl'fɪl

operate= work, run, organize. 'ɒpəreɪt

expert= specialist, professional #amateur 'ekspɜ:t

recreate= to make something from the past exist again in a new form or be experienced again rɪ:kri'eɪt

asset= the things that a company owns, that can be sold to pay debts 'æset

exhibit= something, for example a painting, that is put in a public place so that people can go to see it ɪg'zɪbɪt

be engaged in something= to be doing or to become involved in an activity. bi ɪn'geɪdʒd ɪn 'sʌmθɪŋ

interpret= explain, clarify, make clear. ɪn'tɜ:pɪt

steer a course= to choose a way of doing something carefully. stɪə ə kɔ:s

accuracy must be increasingly **altered**. For example, **Pithecanthropus erectus** is **depicted** in an Indonesian museum with Malay facial features, because this **corresponds** to public **perceptions**. Similarly, in the Museum of Natural History in Washington, Neanderthal man is shown making a **dominant gesture** to his wife. Such presentations tell us more about **contemporary perceptions** of the world than about our **ancestors**. There is one **compensation**, however, for the professionals who make these **interpretations**: if they did not provide the **interpretation**, visitors would do it for themselves, based on their own ideas, **misconceptions** and **prejudices**. And no matter how exciting the result, it would contain a lot more **bias** than the presentations provided by **experts**.

F
Human **bias** is **inevitable**, but another **source** of **bias** in the **representation** of history has to do with the **transitory nature** of the materials themselves. The simple fact is that not everything from history survives the historical process. Castles, palaces and **cathedrals** have a longer **lifespan** than the **dwellings** of ordinary people. The same applies to the furnishings and other **contents** of the **premises**. In a town like Leyden in Holland, which in the seventeenth century was **occupied** by approximately the same number of **inhabitants** as today, people lived within the walled town, an area more than five times smaller

accuracy= correctness, exactness, precision
'ækjərəsi
pithecanthropus erectus= **homo erectus**= an early form of human that was able to walk on two legs
piθi'kanθrəpəs i'rektəs
depict= describe, illustrate, portray. di'pɪkt
correspond to/with something= link, relate, match. kɒrɪ'spɒnd tu/ wɪð 'sʌmθɪŋ
perception= view, opinion, experience. pə'sepʃən
dominant= more powerful, important, or noticeable than other people or things. 'domɪnənt
gesture= a movement of part of your body, especially your hands or head, to show what you mean or how you feel 'dʒestʃə
contemporary= modern, current. kən'tempərəri
ancestor= a member of your family who lived a long time ago 'ænsəstə
compensation= when someone behaves in a particular way in order to replace something that is missing or to balance the bad effects of something
kɒmpən'seɪʃən
interpretation= explanation, understanding, clarification ɪn,tɜ:'pri'teɪʃən
misconception= misunderstanding, mistaken belief, error. mɪskən'sepʃən
prejudice= an unreasonable dislike and distrust of people who are different from you in some way, especially because of their race, sex, religion etc - used to show disapproval 'predʒədɪs
bias= an opinion about whether a person, group, or idea is good or bad that influences how you deal with it. 'baɪəs

inevitable= unavoidable, certain, usual. ɪ'nevətəbel
representation= a painting, sign, description etc that shows something reprɪzen'teɪʃən
transitory= continuing or existing for only a short time 'trænzətəri
nature= the qualities or features that something has 'neɪtʃə
cathedral= church kə'θɪ:drel
lifespan= lifetime, lifecycle, existence 'laɪfspæn
dwelling= house, apartment, flat 'dwelɪŋ
premises= buildings 'premɪsɪz
occupy= live in, inhabit. 'ɒkjəpaɪ
inhabitant= resident, citizen, occupant ɪn'hæbɪtənt

than modern Leyden. In most of the houses several families lived together **in circumstances** beyond our imagination. Yet in museums, **fine** **period** rooms give only an image of the lifestyle of the **upper class** of that **era**. **No wonder that** people who **stroll** around exhibitions are **filled with nostalgia**; the evidence in museums indicates that life was so much better in the past. This **notion** is **induced** by the **bias** in its **representation** in museums and heritage centres.

in circumstances the conditions that affect a situation, action, event etc in 'sɜ:kəmstənsɪz
fine= attractive, neat, and delicate faɪn
period= having a style typical of a particular time in history 'pɪəriəd
upper class= the group of people who belong to the highest social class 'ʌpə klɑ:s
era= period, age, epoch. 'ɪərə
(it's) no/small/little wonder (that)= used to say that you are not surprised by something ('ɪts) 'nəʊ/smɔ:l/ 'lɪtl/ 'wʌndə (ðæt)
stroll= leisurely walk, wander. strəʊl
be filled with admiration/joy/happiness etc= if you are filled with an emotion, or if it fills you, you feel it very strongly bi fiɪld wɪð ,ædmə'reɪʃn/ 'dʒɔɪ/ 'hæpɪnəs et'setə
nostalgia= a feeling that a time in the past was good, or the activity of remembering a good time in the past and wishing that things had not changed. nɒ'stældʒə
notion = idea, view, concept, belief, perception, thought. 'nəʊʃən
induce= cause, produce, provoke ɪn'dju:s